Huawei Y9 Prime 2019 Schematic Diagram

Ayasofya Grand Mosque 24/7/2020

Ayasofya-i Kebir Camii Şerifi 24/7/2020

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HUAWEI

Schematic Diagram





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Baseband Schematic

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MODEM Schematic

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80. ANT NFC
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83. ANT DIV Tuner
84. ANT GPS WIFI
85. ANT MIMO

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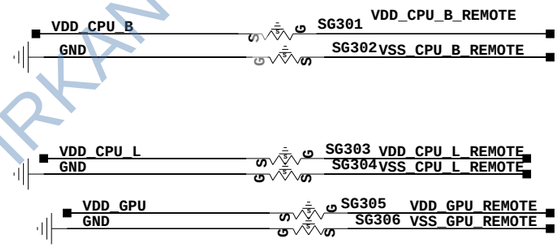
2.Block Diagramm

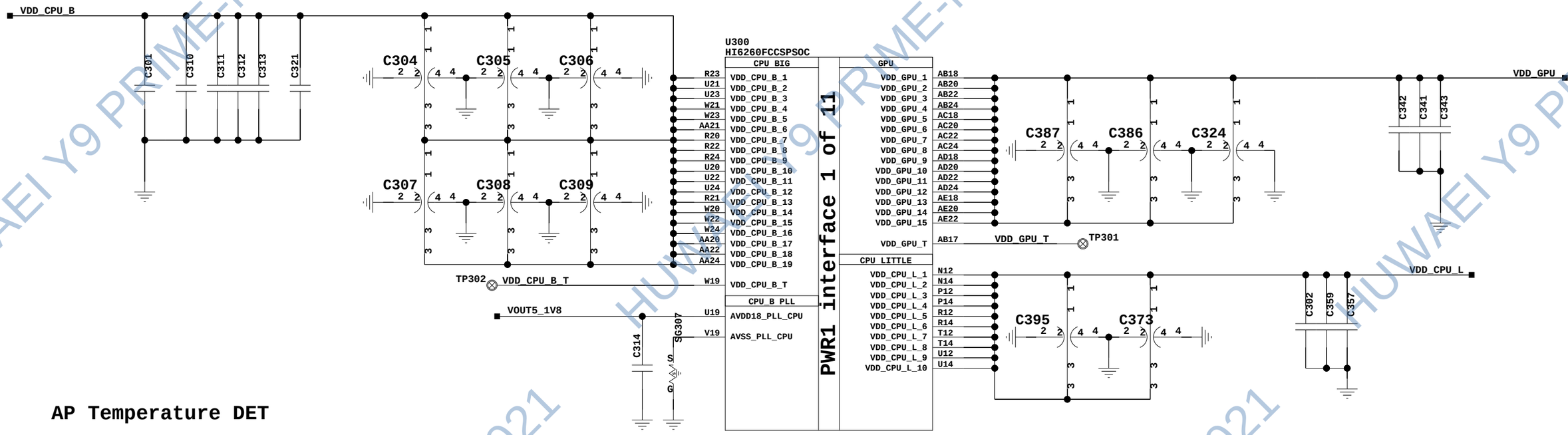
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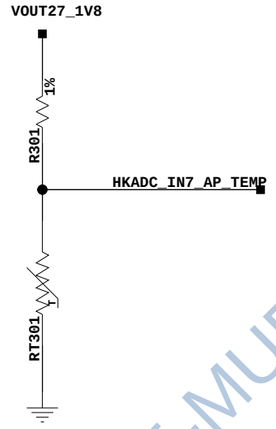
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3.SOC PWR1





AP Temperature DET

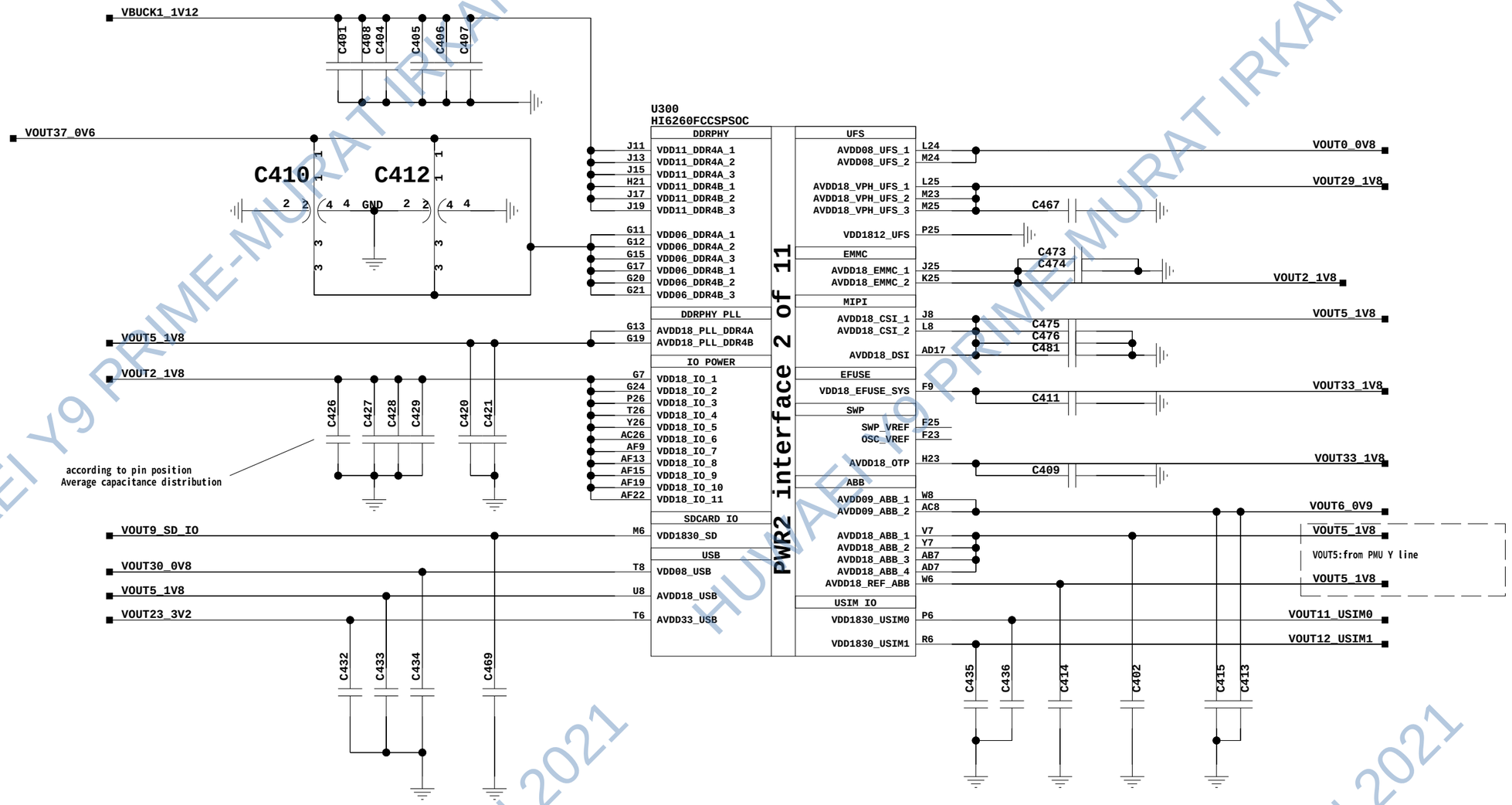


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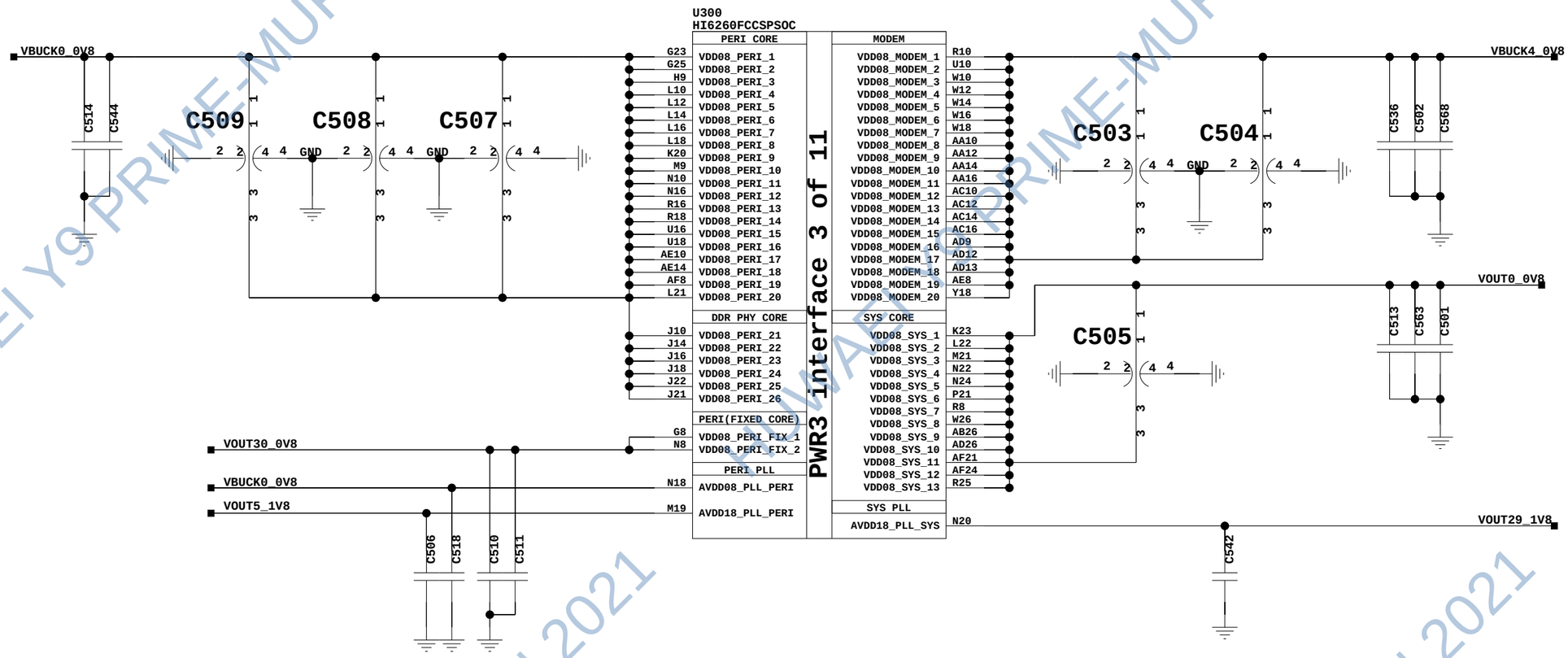
4.SOC PWR2



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5.SOC PWR3

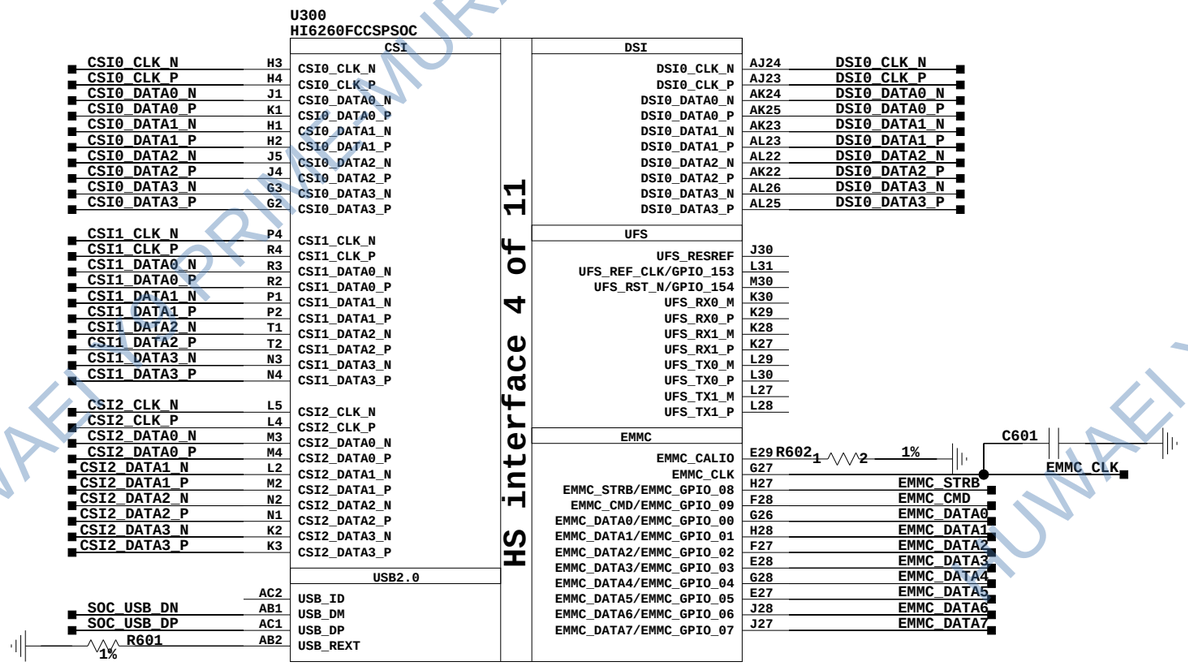


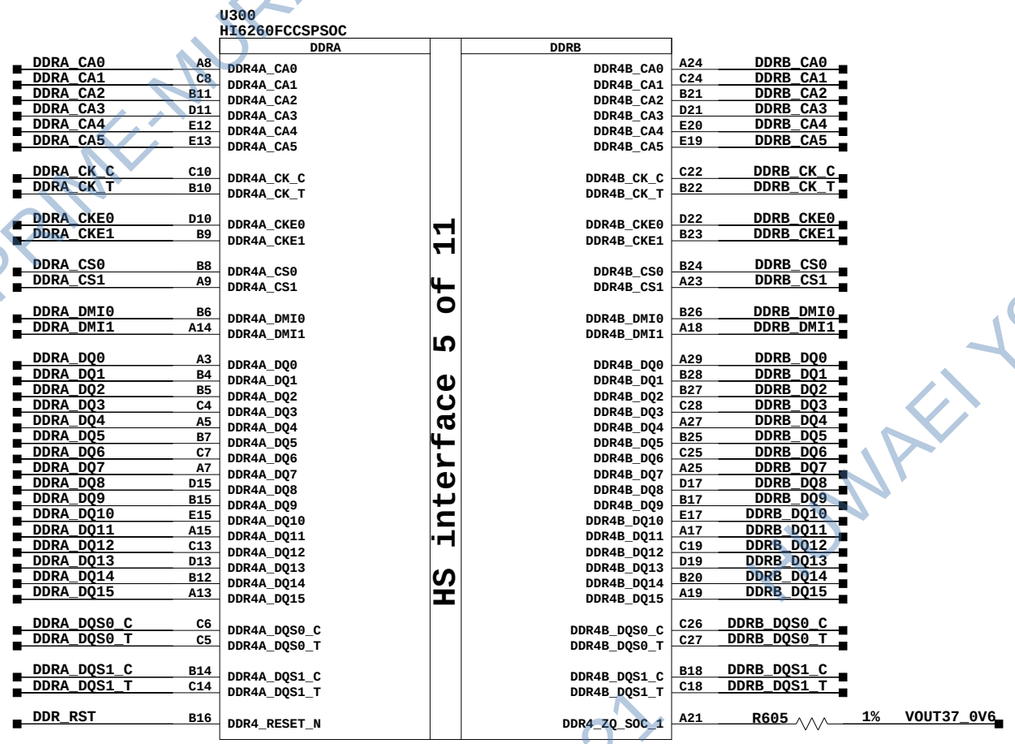
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6.SOC HS Interface





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7.SOC GPIO Interface



GPIO Not Used Is marked ***,Used For CAD to Fanout Net

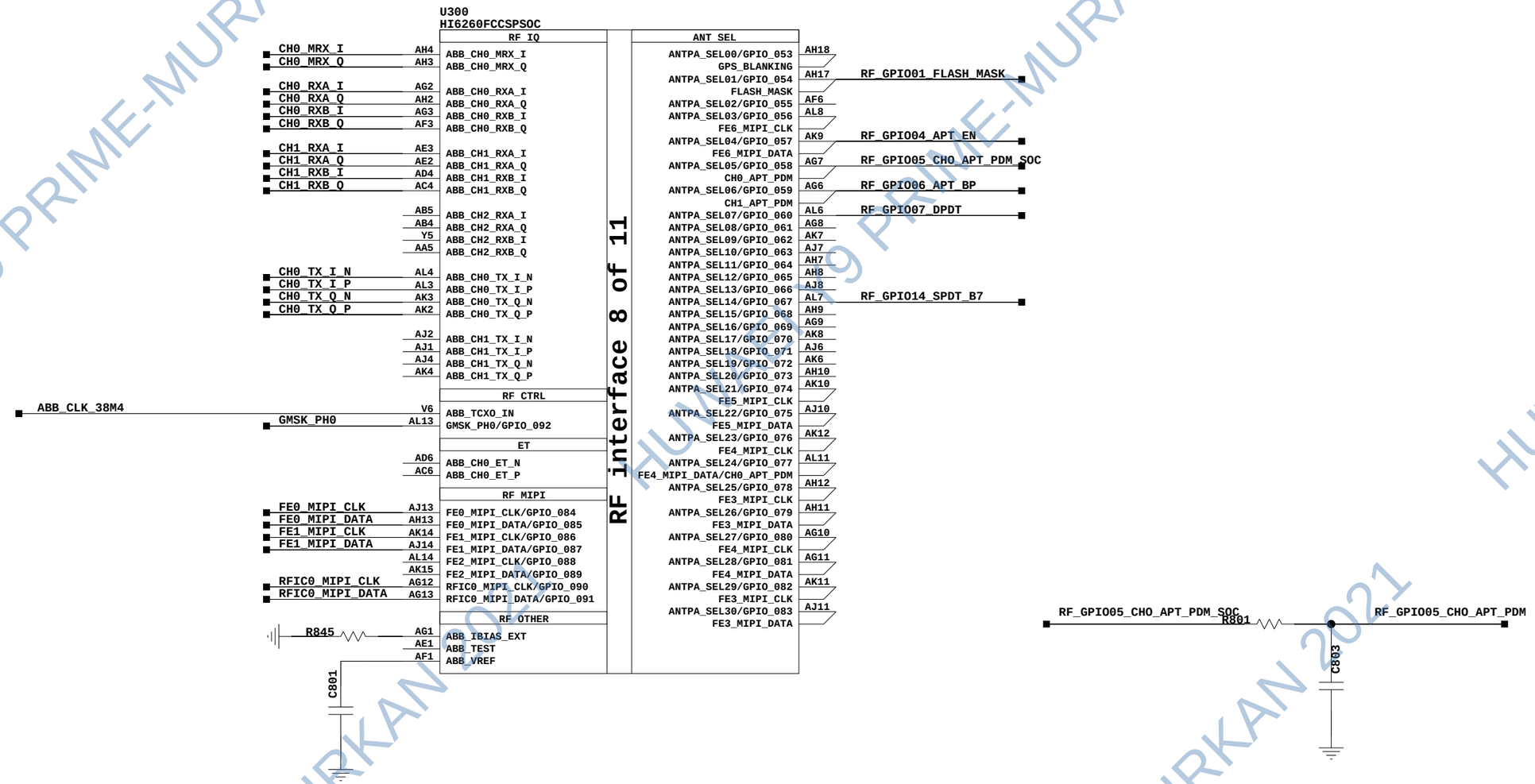
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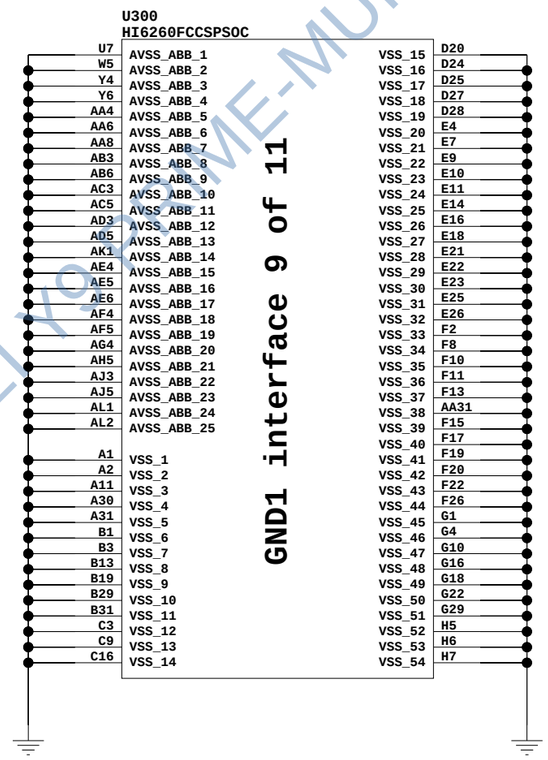
8.SOC RF Interface

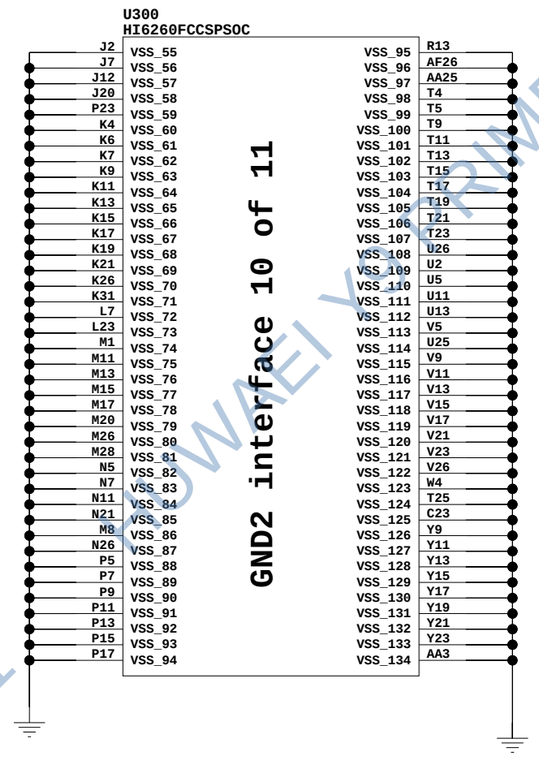
RF IO Not Used Is marked ***,Used For CAD to Fanout Net

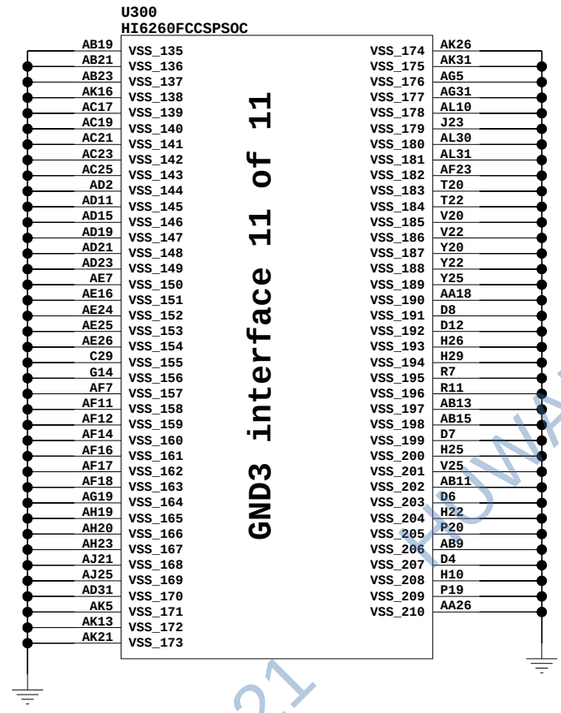


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9.SOC GND





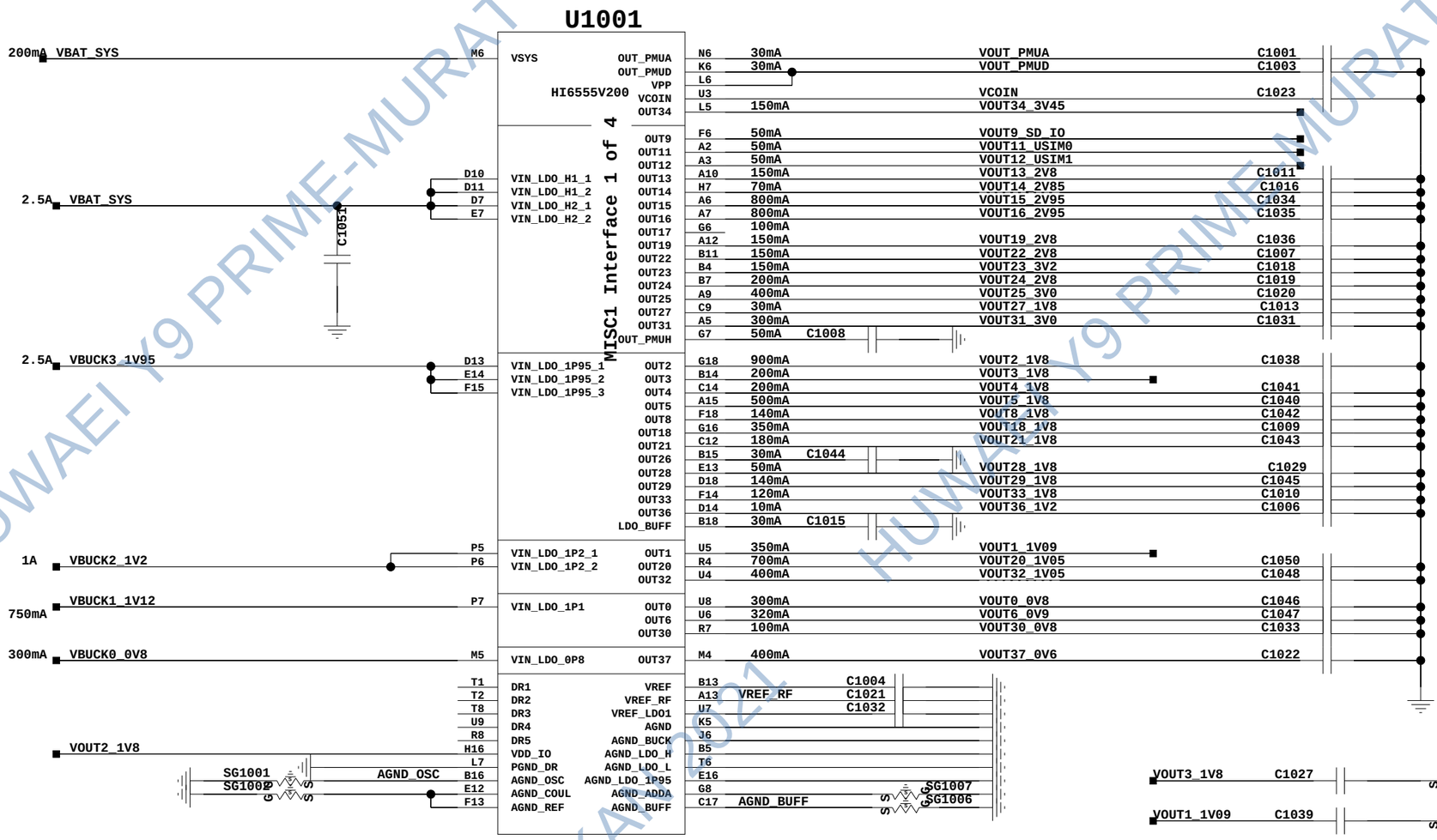


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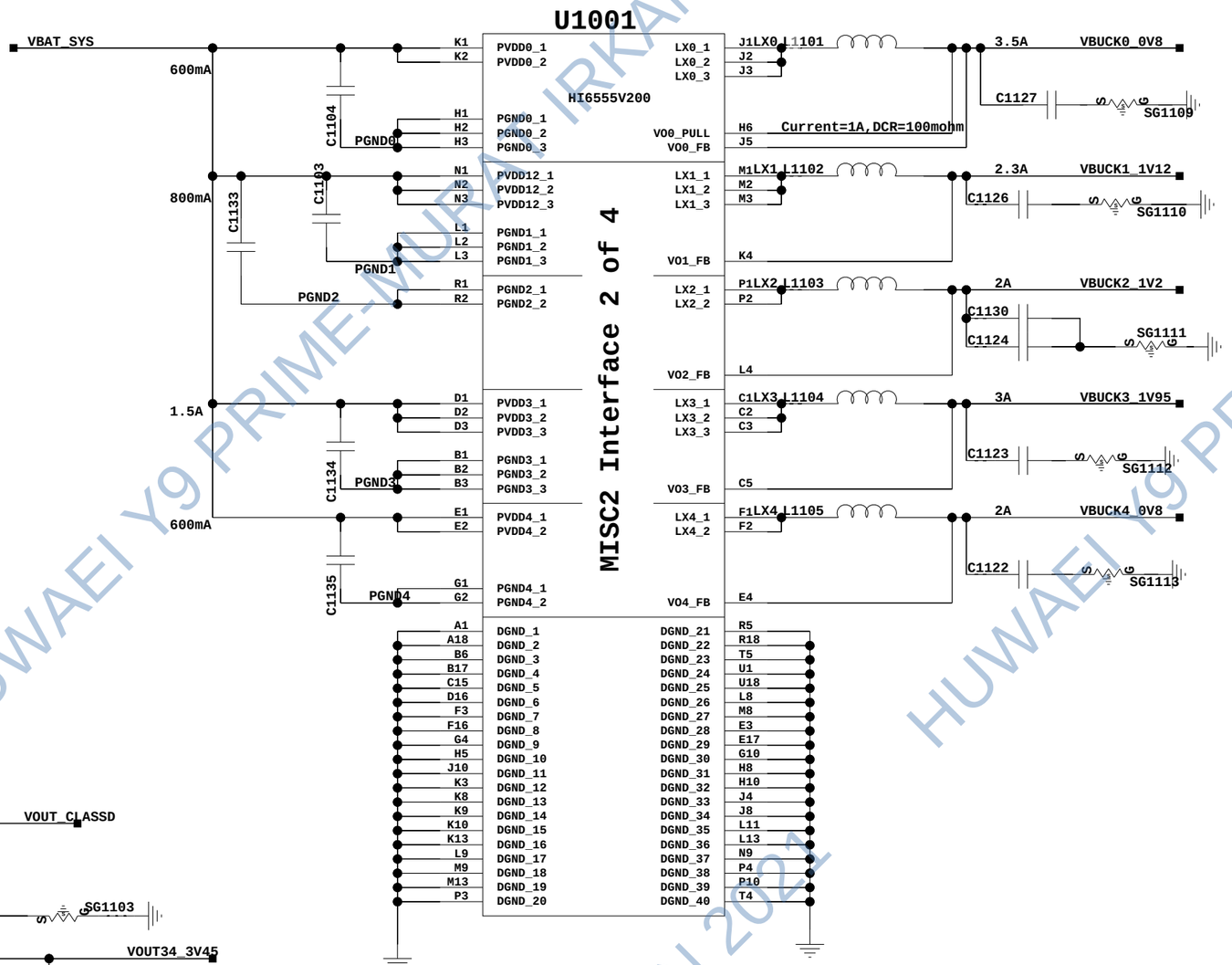
10.PMU LDO

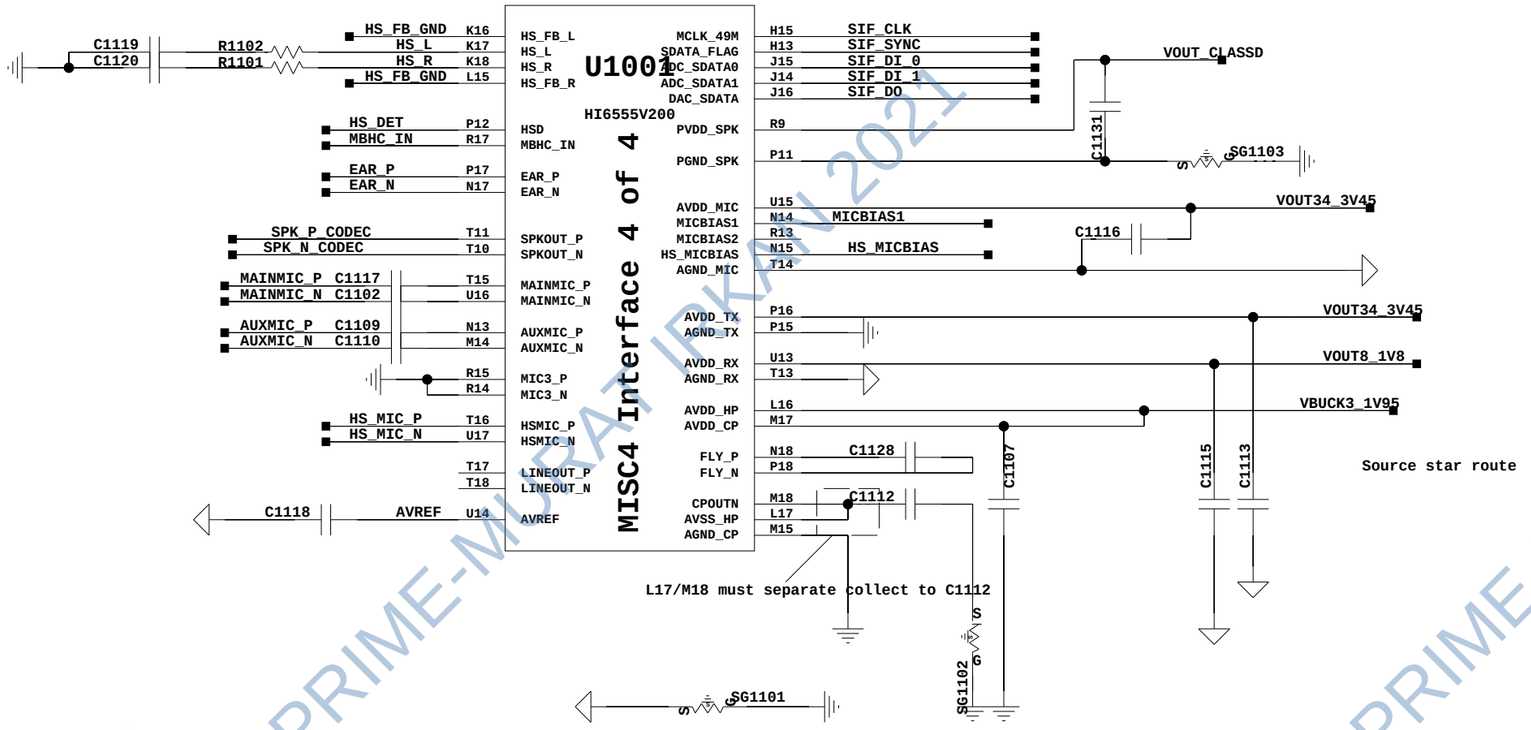


Output	default	current	TYP voltage	function
LDO1	OFF	350	1.09	RFIC0_AVDD
LDO2	ON	900	1.8	PMU dig/RFIC/SOC1V8IO/64031V8/LPDDR4X VDD/UFS
LDO3	OFF	200	1.8	RFIC0_AVDD2
LDO4	OFF	200	1.8	LCD1V8 IO/TP1V8 IO
LDO5	ON	500	1.8	PHY ANA/ABB_AVDD1V8
LDO8	OFF	140	1.8	6403ANALOG/CLASSD1V8
LDO9	OFF	50	1.8/2.95	SD IO
LDO11	OFF	50	1.8	SIM0
LDO12	OFF	50	1.8	SIM1
LDO13	OFF	150	2.85	S camera analog
LDO14	OFF	70	2.85	RF FEM/AUXDAC
LDO15	ON	800	2.95	EMMC/UFS VDD
LDO16	OFF	800	2.95	SD card
LDO17	OFF	100	2.8	LCD TP VCI
LDO18	ON	350	1.8	fp/SENSOR IO FP AVDD/CON IO/NFC IO
LDO19	OFF	300	2.8	M camera anlog
LDO20	OFF	400	1.11	M Camera Core
LDO21	OFF	180	1.8	Camera io
LDO22	OFF	150	2.8	Back Camera Analog
LDO23	ON	150	3.2	HKADC
LDO24	ON	200	2.8	X Sensor AVDD
LDO25	OFF	400	3.0	Camera VCM/mipi switch
LDO26	ON	30	1.35	38.4_AVDD
LDO27	ON	50	1.8	HKADC
LDO28	OFF	50	1.8	RF MIPI I/O
LDO29	ON	140	1.8	UFS PHY/SYS CLK_BUF
LDO30				
LDO31	OFF	300	3.0	Vibrator
LDO32	OFF	400	1.1	S Camera Core
LDO33	OFF	120	1.8	EFUSE
LDO34	ON	150	3.45	CODEC RCV/MICBIAS

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11.PMU BUCK/CODEC



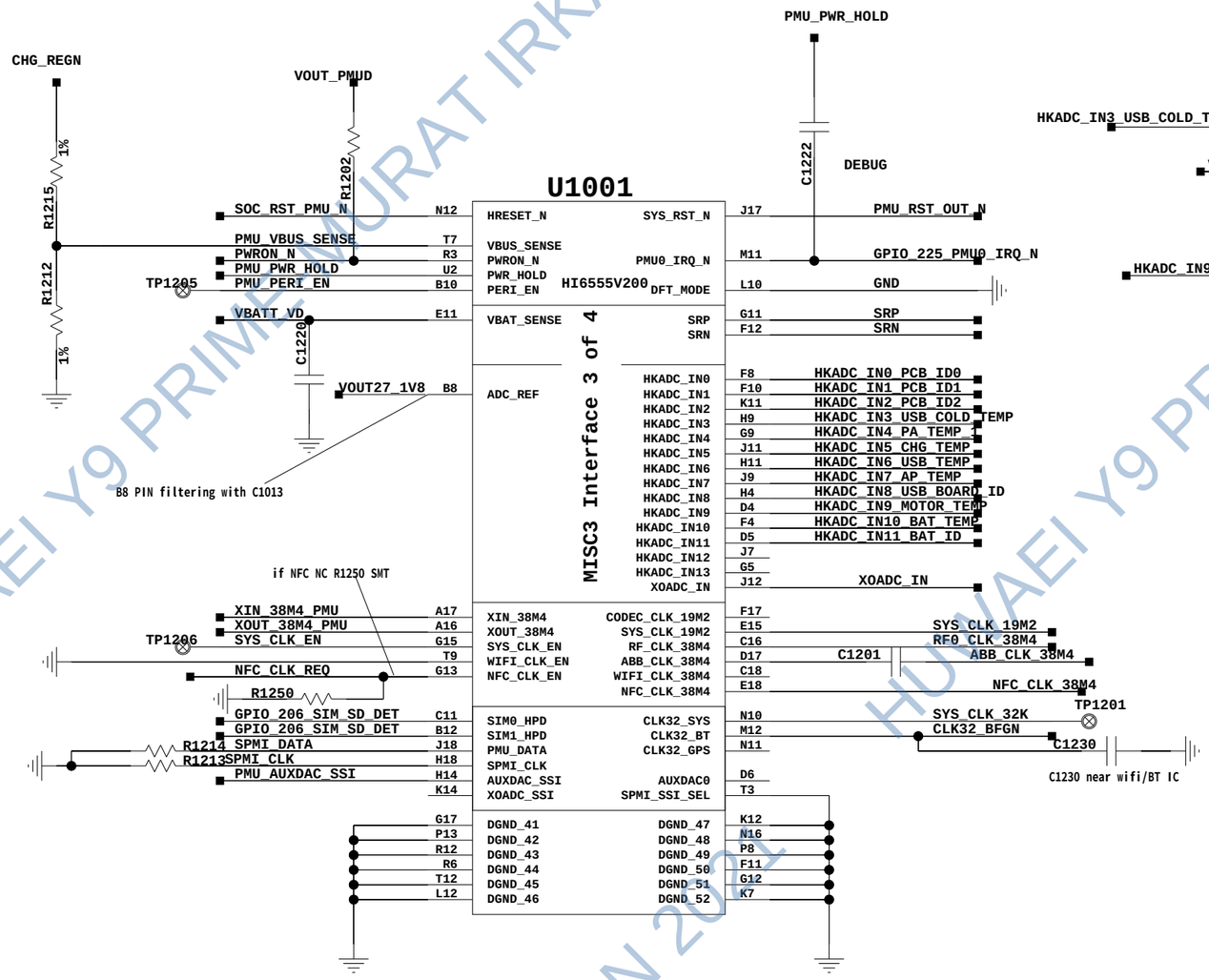


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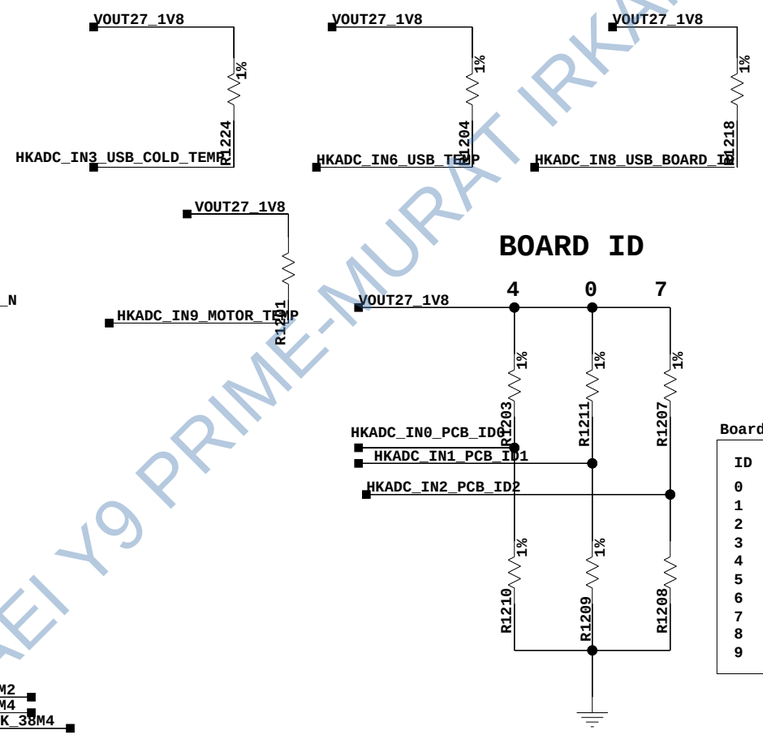
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12. PMU Digital



USB CON TEMP DETUSB Board ID DET



HKADC·ÖÃä±í

HKADC_IN0	APU Board ID0(RF USE)
HKADC_IN1	APU Board ID1
HKADC_IN2	APU Board ID2
HKADC_IN3	USB_COLD
HKADC_IN4	PA TEMP DET0
HKADC_IN5	Charge TEMP
HKADC_IN6	USB TEMP ADC
HKADC_IN7	AP TEMP
HKADC_IN8	USB_BOARD_ID
HKADC_IN9	MOTOR_TEMP
HKADC_IN10	BATT TEMP
HKADC_IN11	BATT ID
HKADC_IN12	RESERVERD
HKADC_IN13	RESERVERD

R1218	Code	Value
STK-L21A/L01A/L21AME/L22B	07091406	200k
STK-L23B/L03B/L22B	07092138	150k
STK-AL00/TL00	07092918	100k
Reserved	07091302	51k
Reserved	07091408	68k
Reserved	07091299	30k
Reserved	07091172	22K
Reserved	TBD	TBD

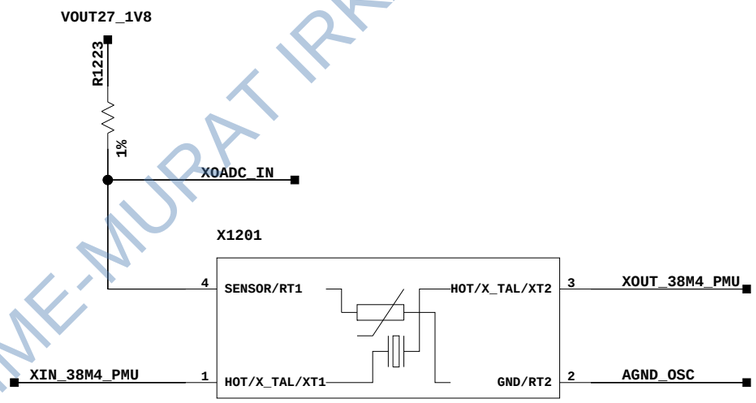
Board ID Resistor Set

ID	UP RES	DOWN RES
0	NC	0K
1	150K	22K
2	100K	30K
3	200K	100K
4	150K	121K
5	121K	150K
6	51K	100K
7	30K	100K
8	20K	150K
9	150K	NC

Resistor Code

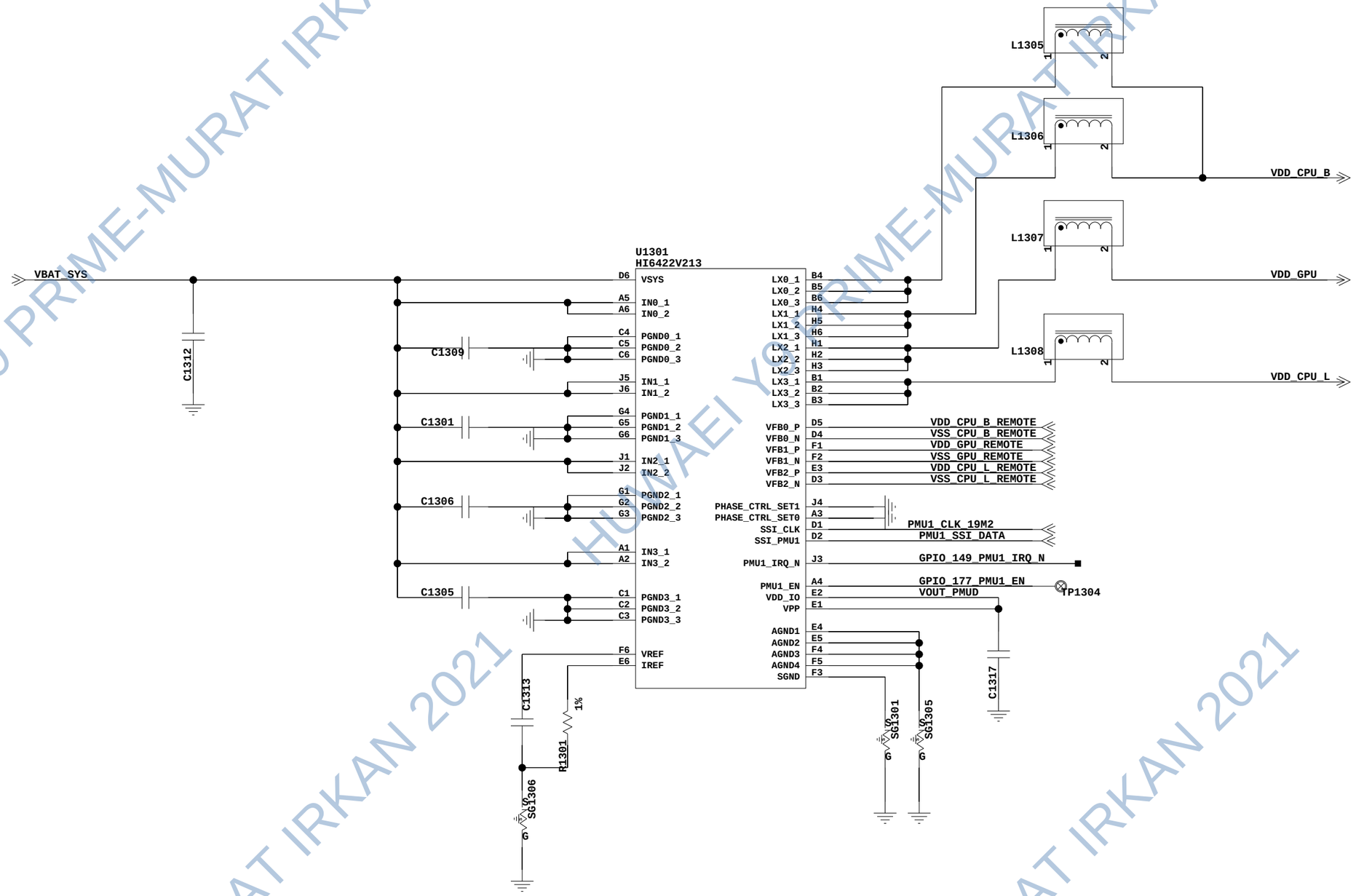
0K	07092922
20K	07092448
22K	07091172
30K	07091299
51K	07091302
100K	07092918
121K	07091176
150K	07092138
200K	07091406

Mode	boardID	R1207	R1208	R1211	R1209	R1203	R1210	
L21A/L01A	7700	30K	100K	NC	0	NC	0K	V0
L21AME	7701	30K	100K	NC	0	150K	22K	V3
L22B	7702	30K	100K	NC	0	100K	30K	V3&VN1
L23B/L03B	7703	30K	100K	NC	0	200K	100K	V3&VN1
L21A/L01A	7704	30K	100K	NC	0	150K	121K	V3&V3.1&VN1
L21AME/BME	7708	30K	100K	NC	0	20K	150K	VN1



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13.Hi6422



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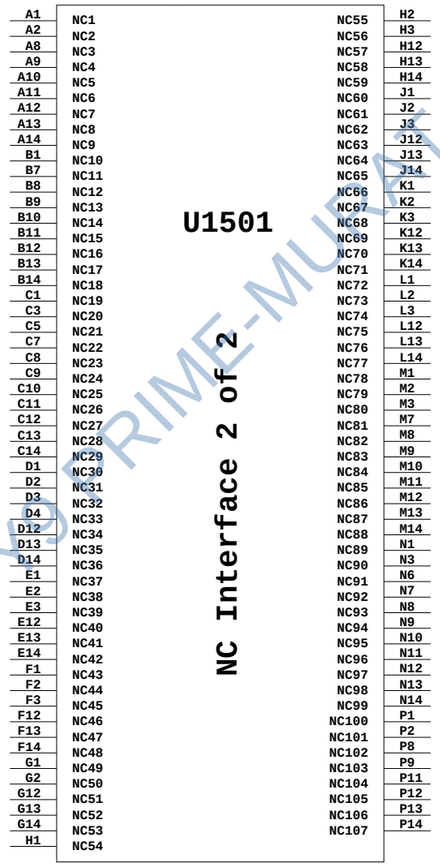
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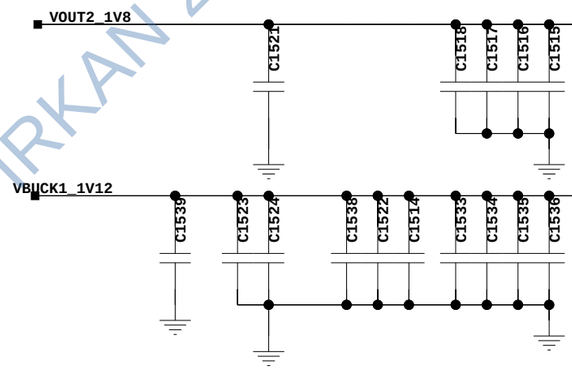
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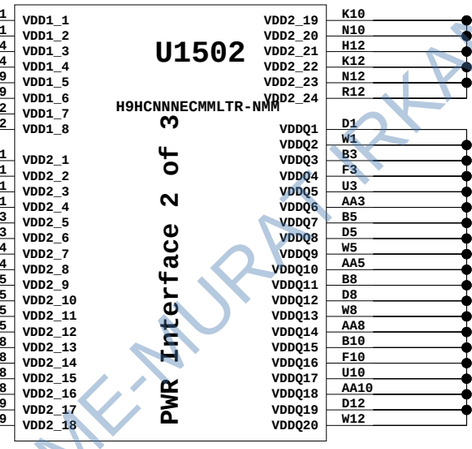
14. Hi6422 Reserved

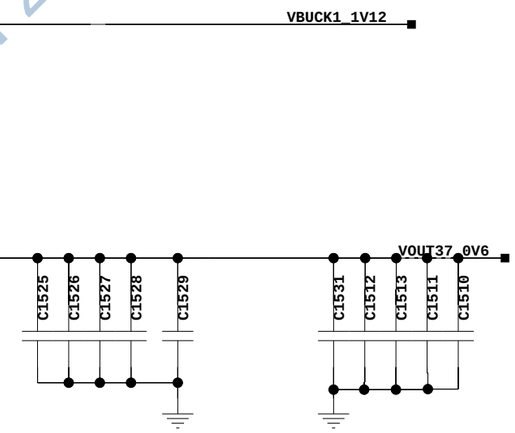
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15. EMMC DDR

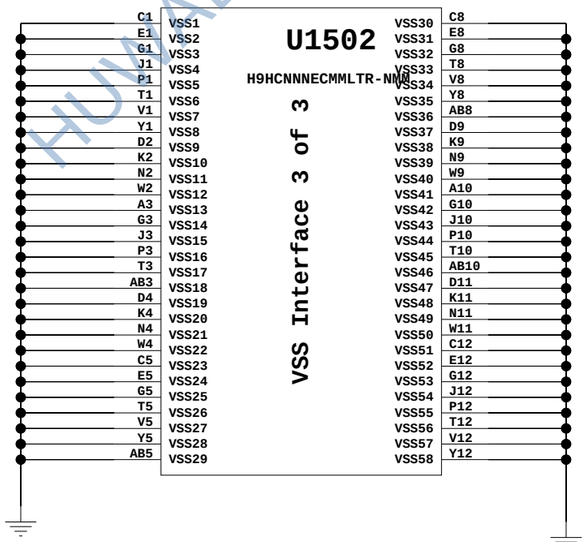


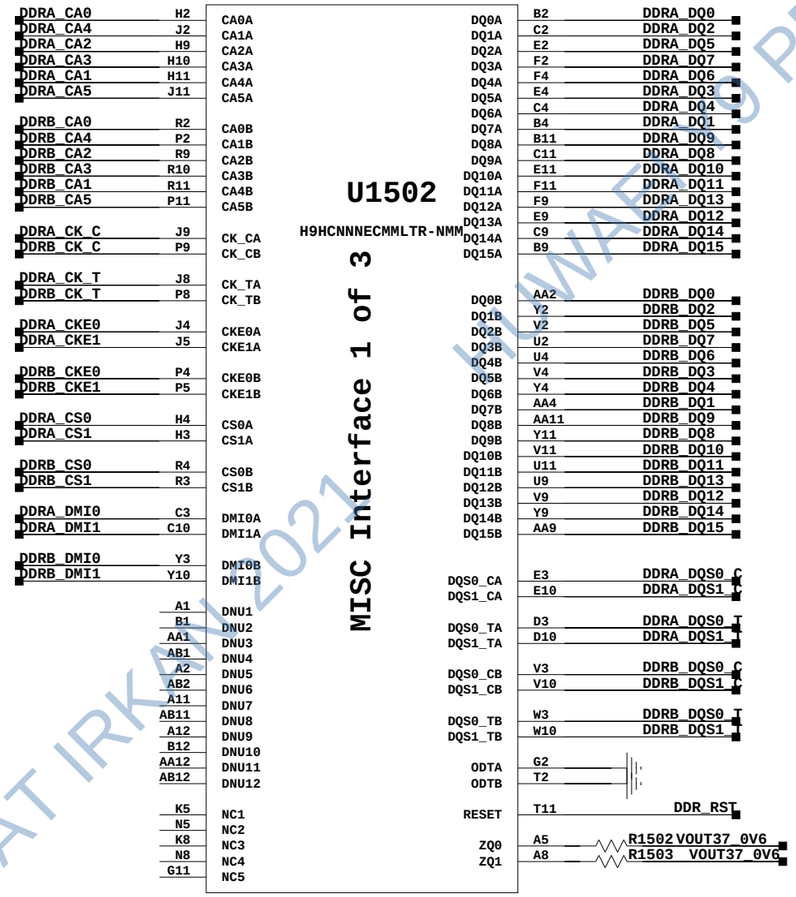


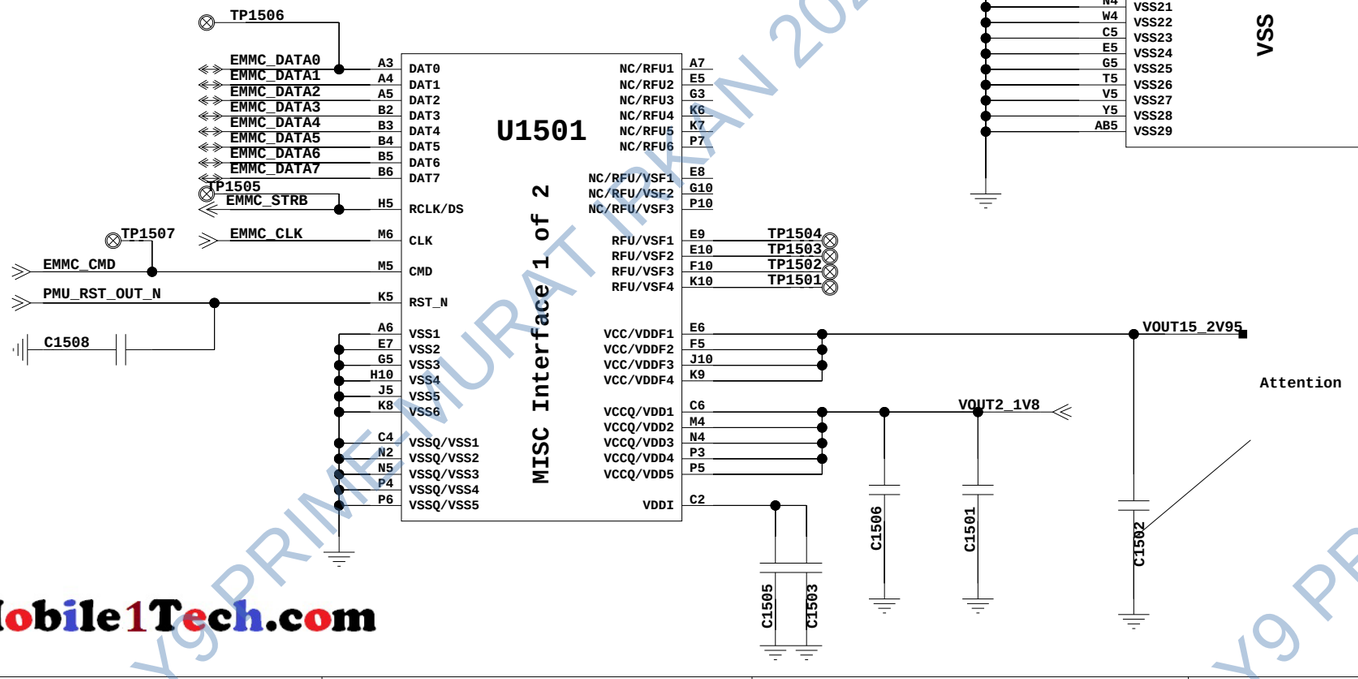




DDR	EMMC
4G--40020556	64G---40061006
6G--40020570	128G--40061007







Attention

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16. UFS/DDR Reserved

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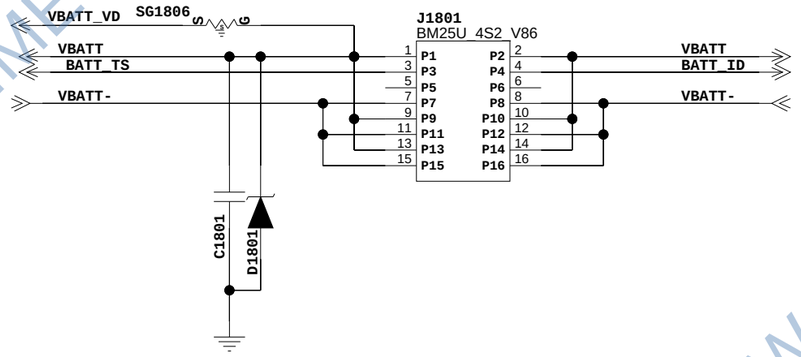
17. External Codec Reserved

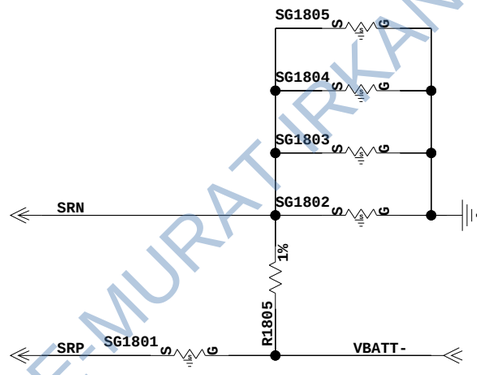
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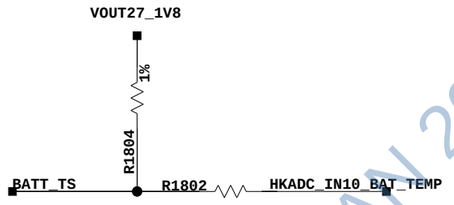
18. Battery & Fuel guage

Battery Connector

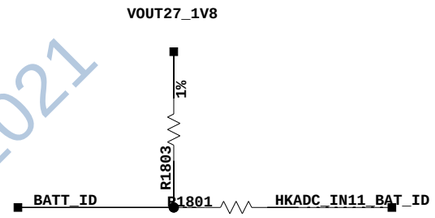




CAD note:1.Trace for 6A
2.Differential trace



Battery Temperature

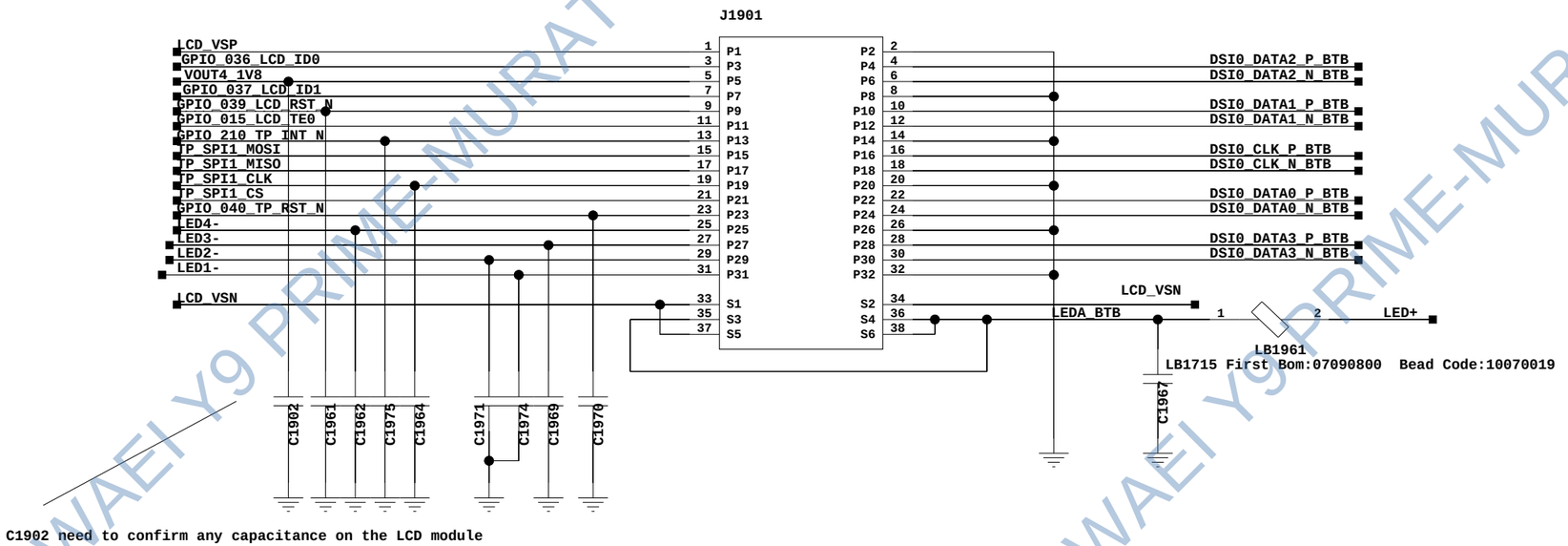


Battery ID

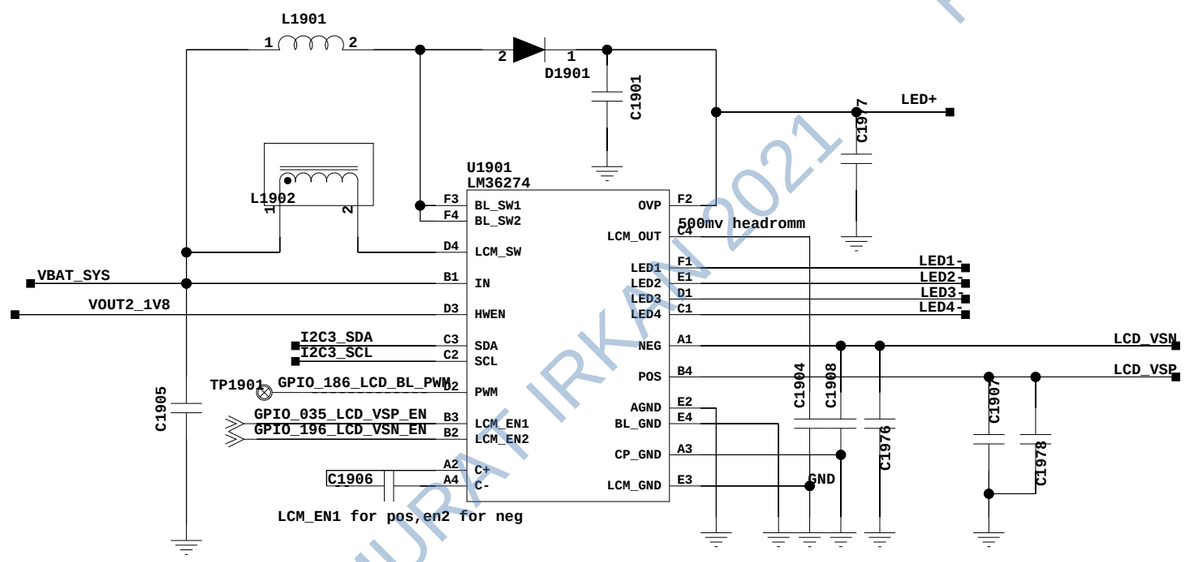
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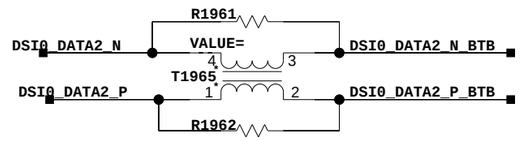
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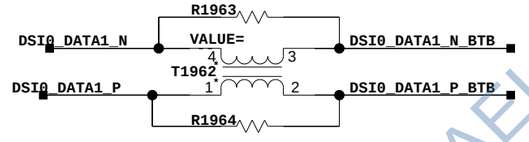
19. LCD and TP INTERFACE

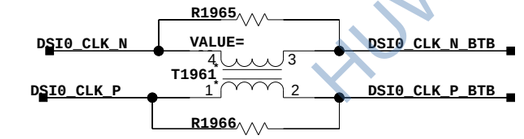


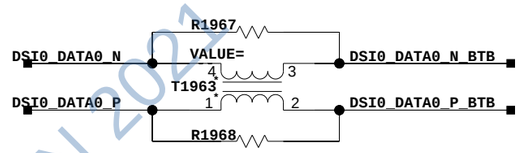
C1902 need to confirm any capacitance on the LCD module

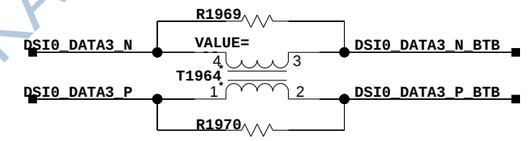












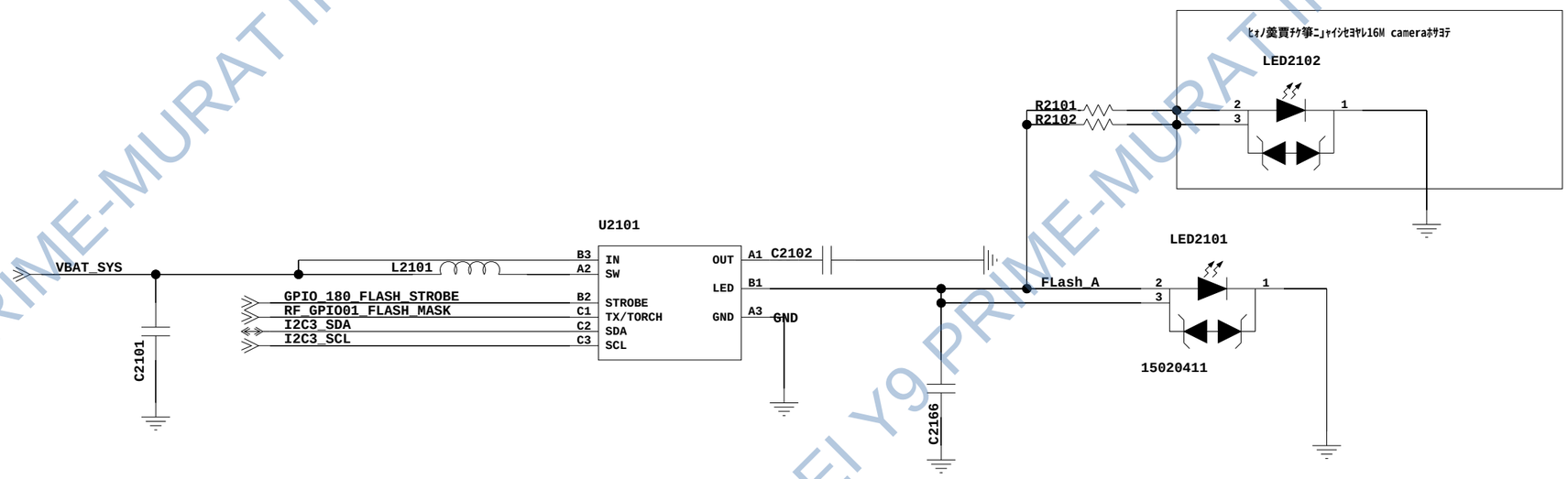
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20. TP/LCD Reserved

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21. Flash/CAM LDO



	2 camera	3 camera
LED2102,R2101,R2102	SMT	DNI
LED2101	DNI	SMT

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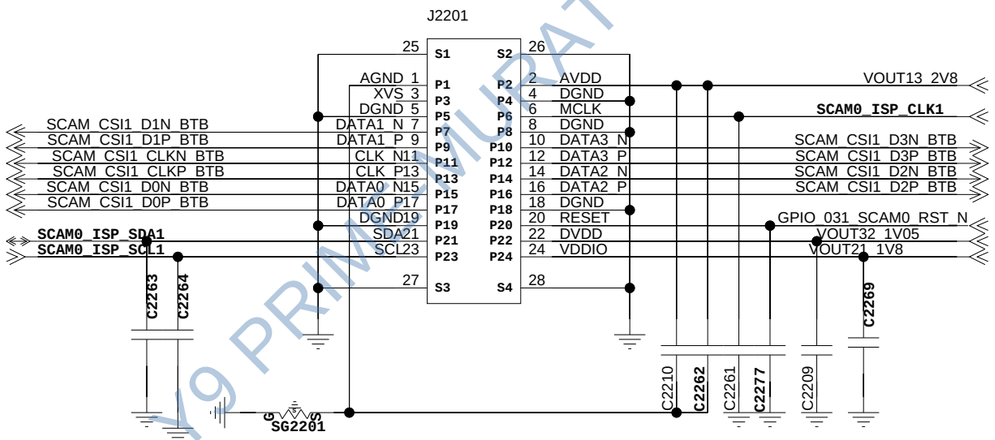
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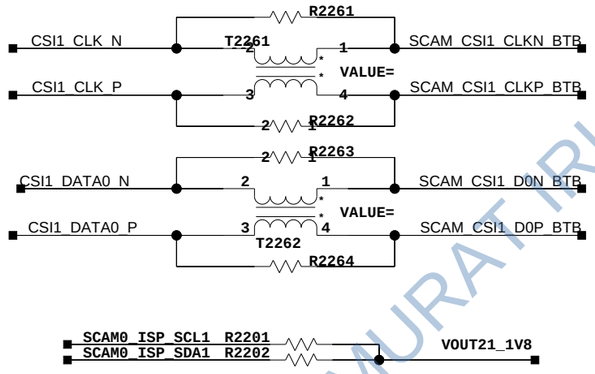
22. Front Camera/MIPI SW

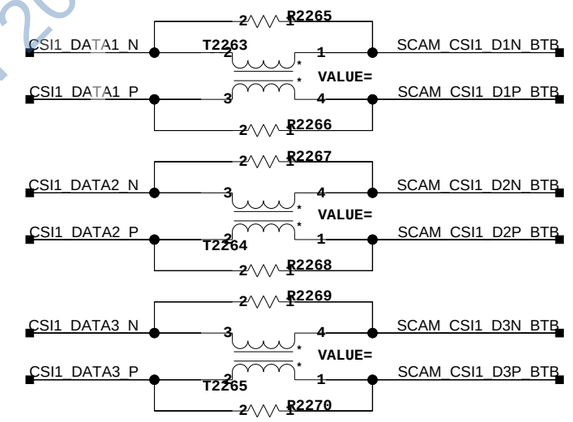
Front SUB Camera

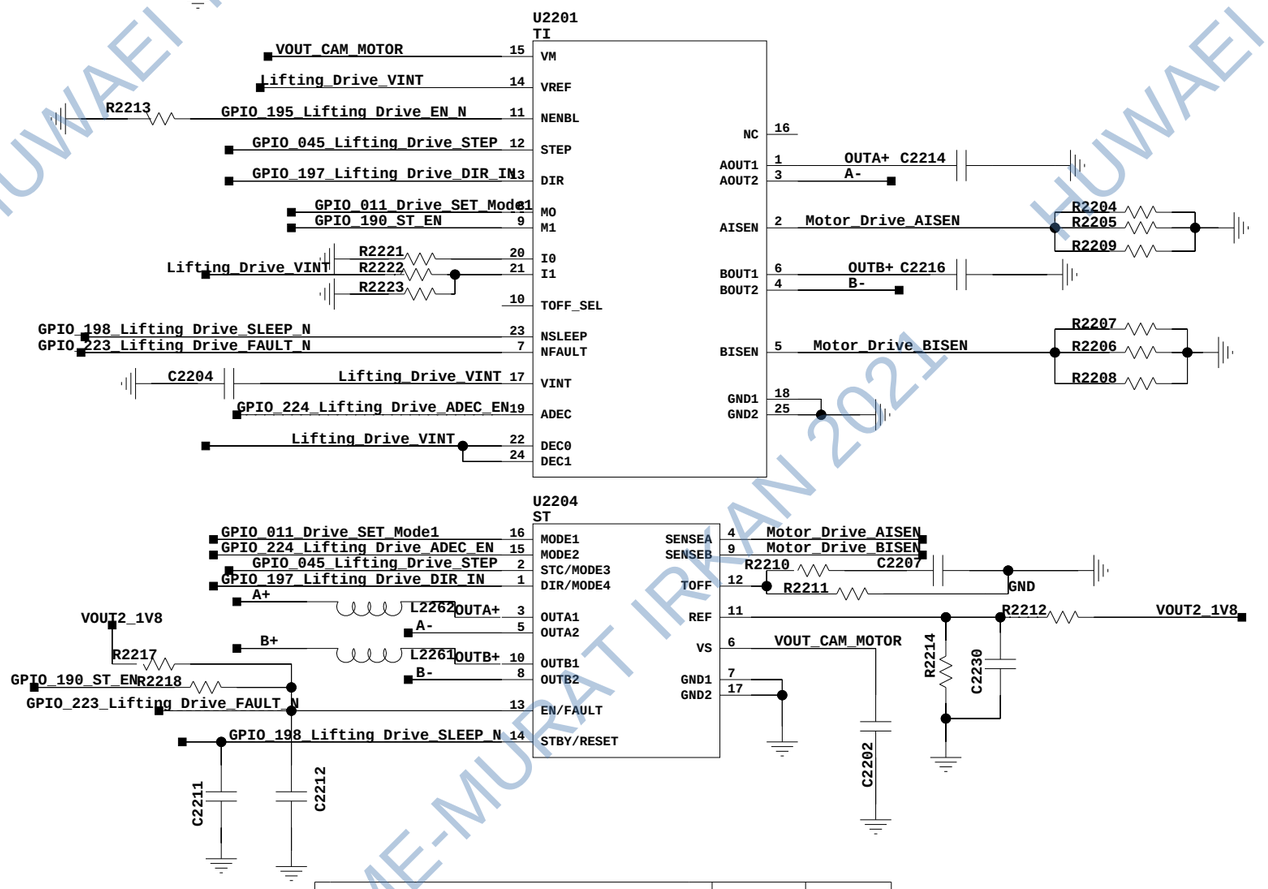
16M Sub 23060300

Max Mated High=0.8mm

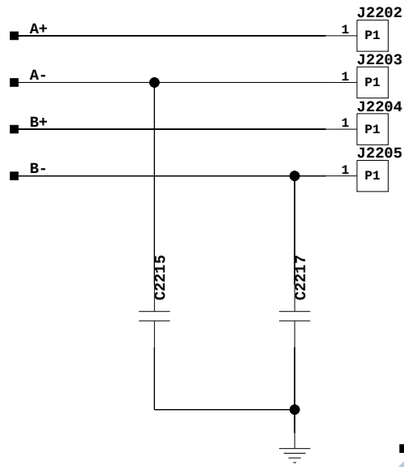


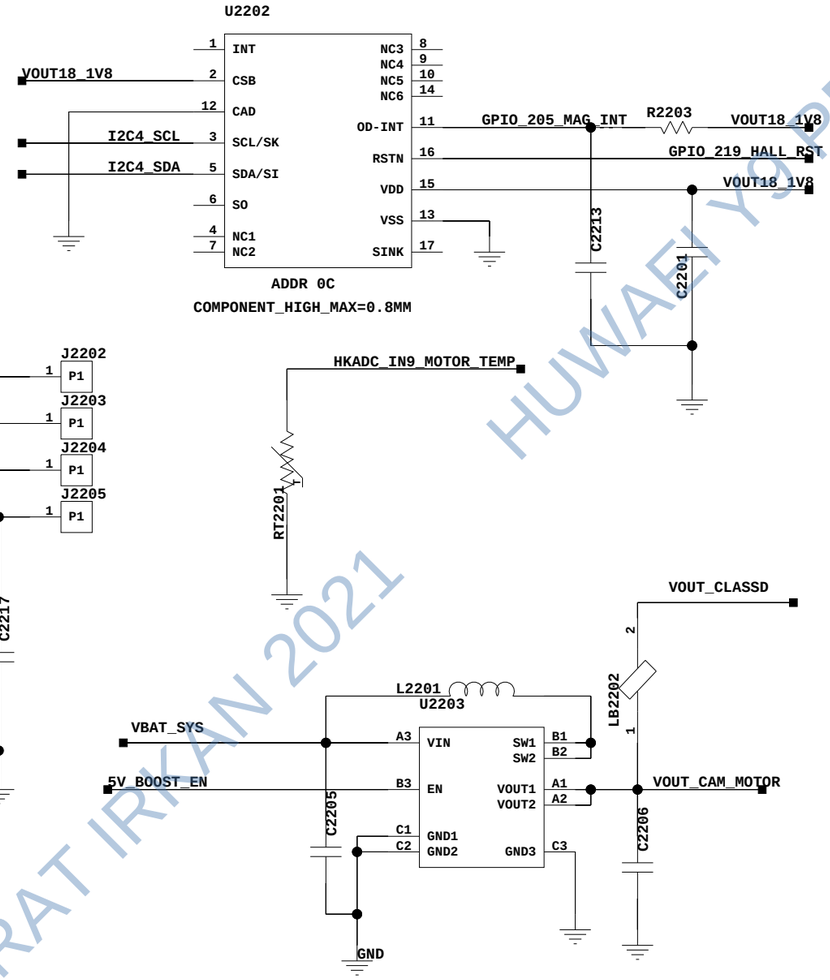






	ST-U2204	TI-U2201
C2211,C2212,C2207,C2230,R2210,R2211,R2212,R2214	SMT	DNI
R2213 R2218R2209 R2208	SMT	DNI
C2204 R2222	DNI	SMT





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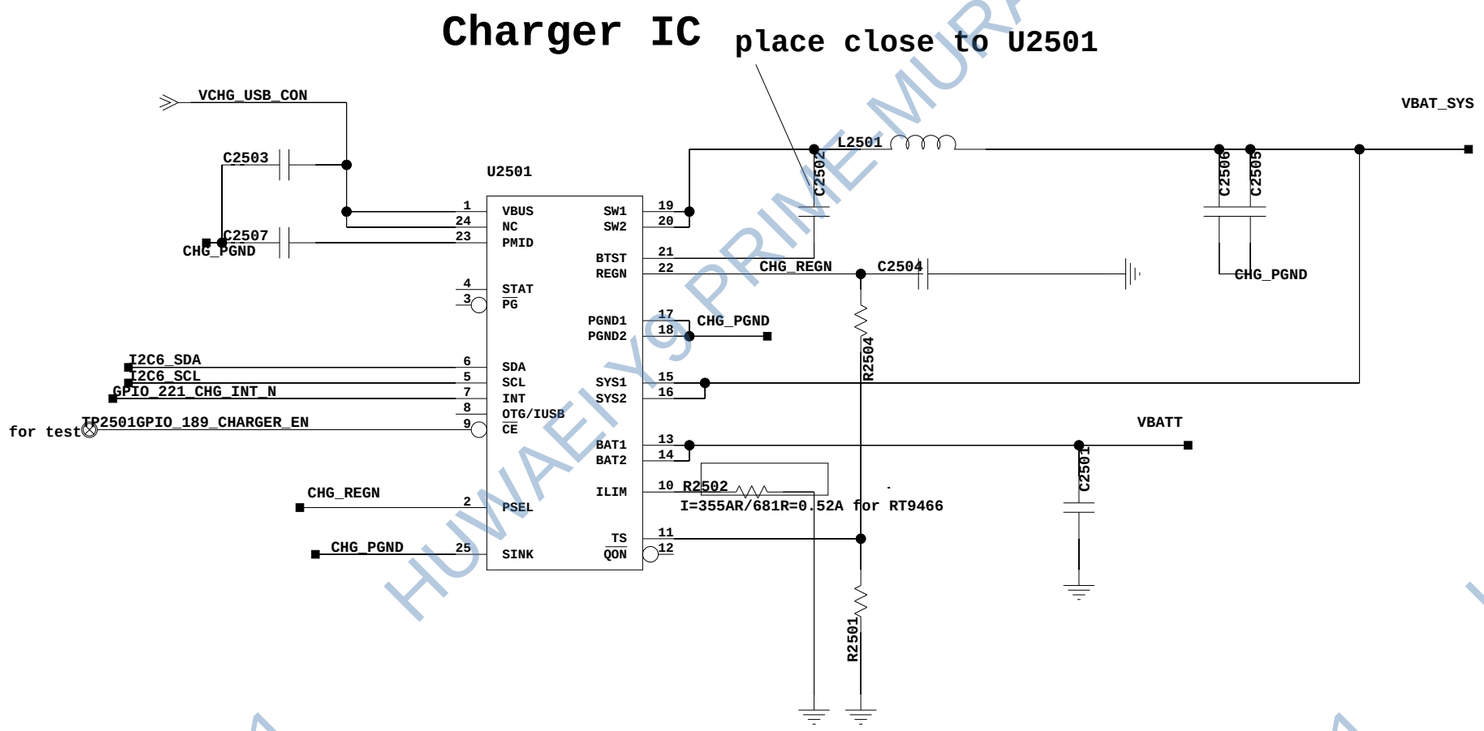
24. CAM Reserved

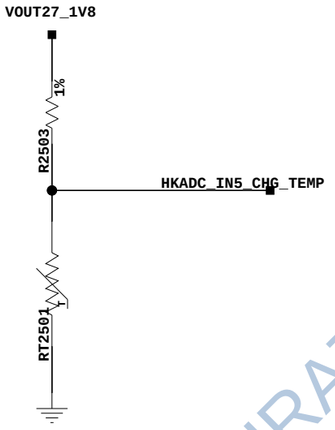
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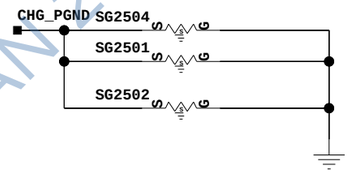
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25. Charge Management







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26. Direct Charger Reserved

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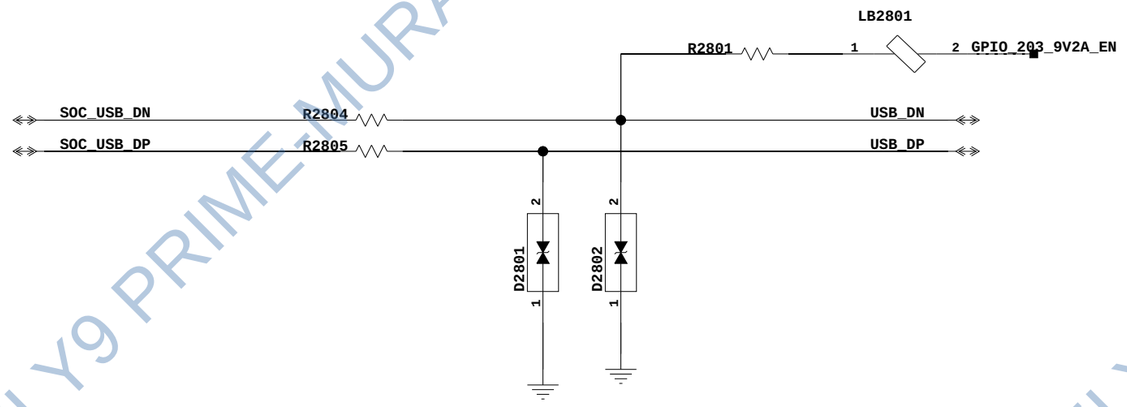
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27. Wireless Charger Reserved

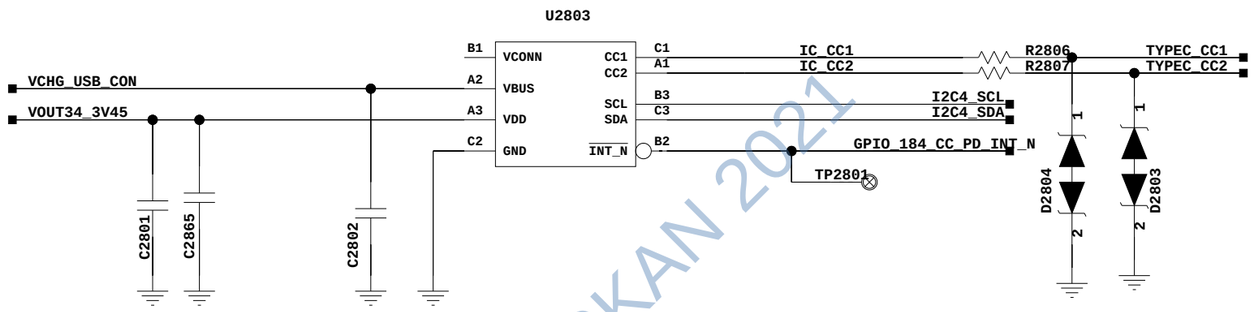
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28. USB Interface



PD



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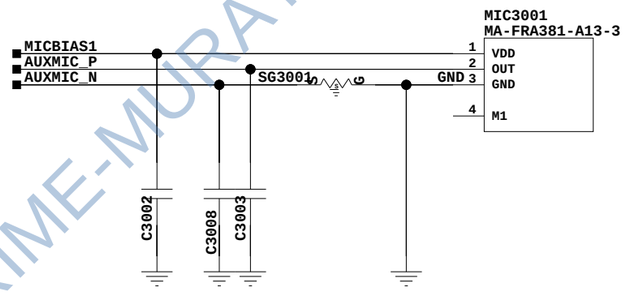
29. USB Interface Reserved

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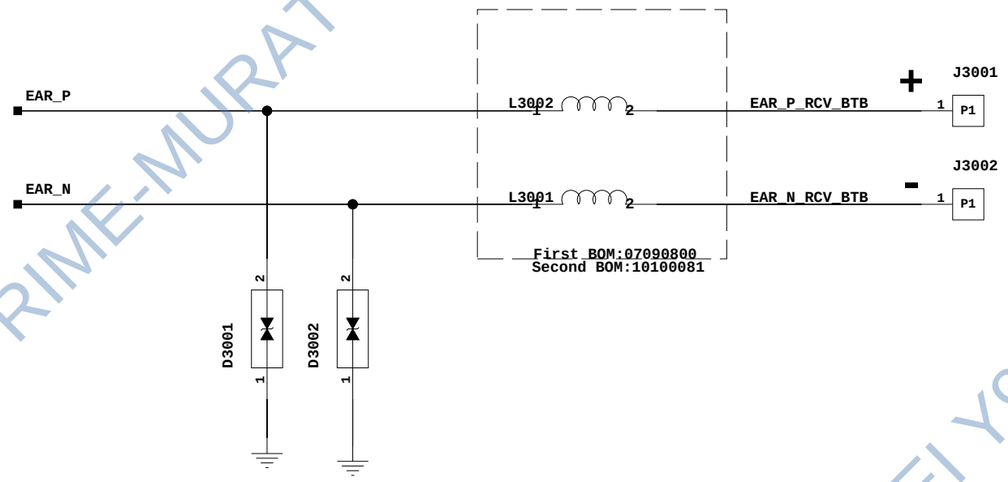
30. MIC/RCV

Second MIC

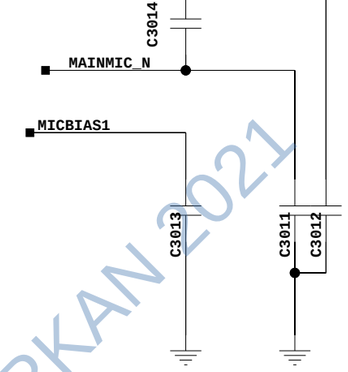


CAD note: 1,MIC need differential trace
2, MIC GND pad need use cycle via to GND net.

RCV



near PMU to placement

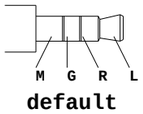


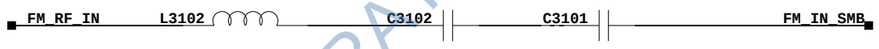
HUWAEI Y9 PRIME-MURAT IRKAN 2021

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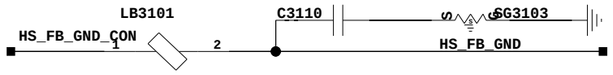
Ayasofya-i Kebir Camii Şerifi 24/7/2020

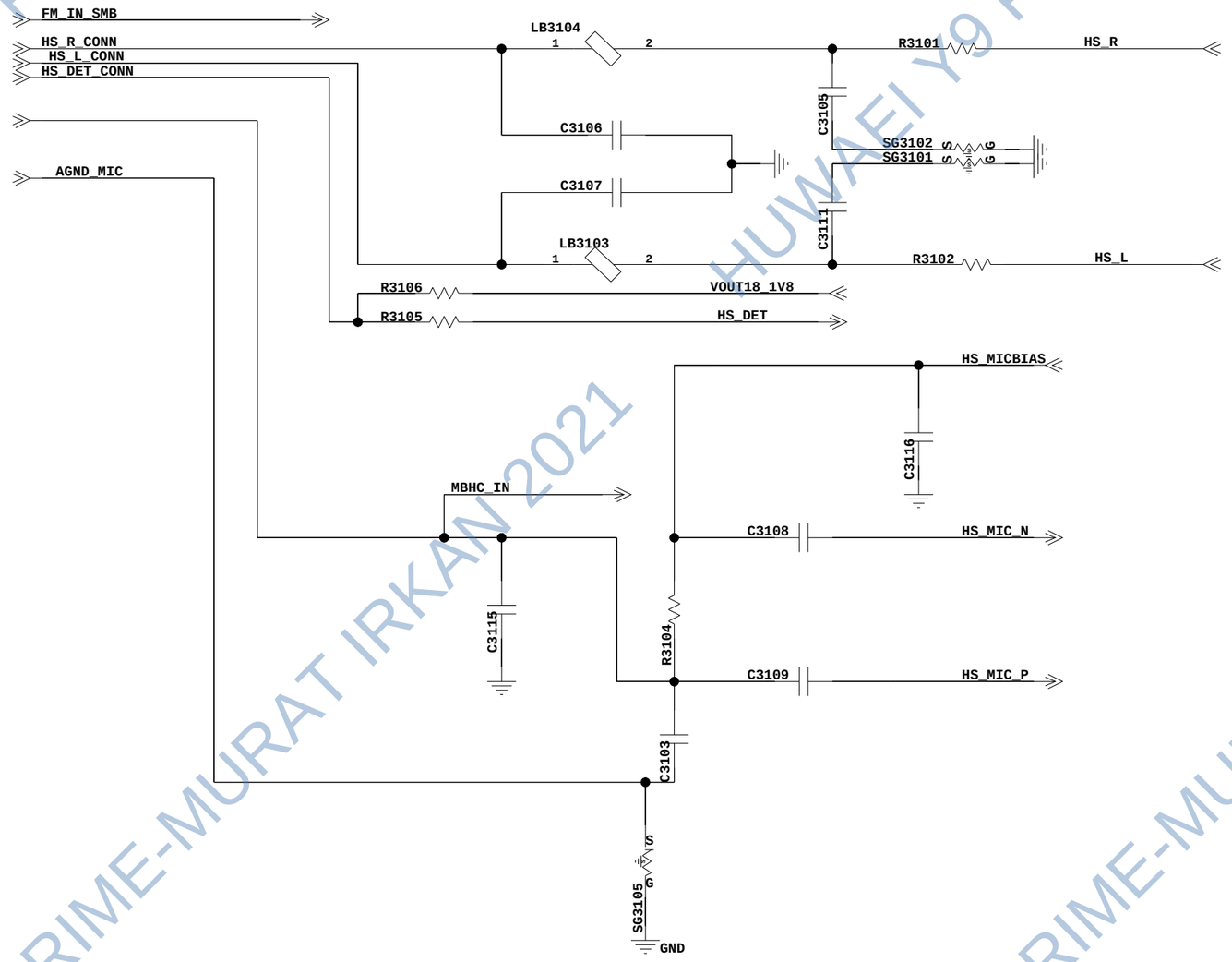
31. Headphone





CLOSE TO Hi1102 IC

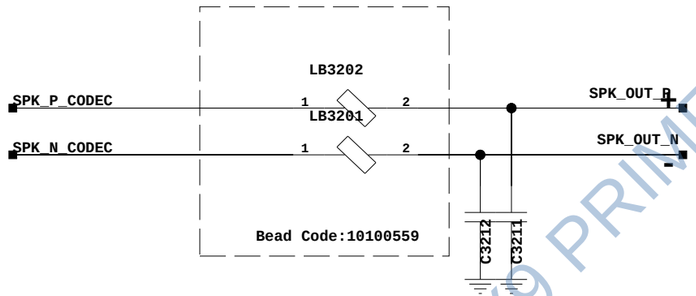




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32. Smart_PA



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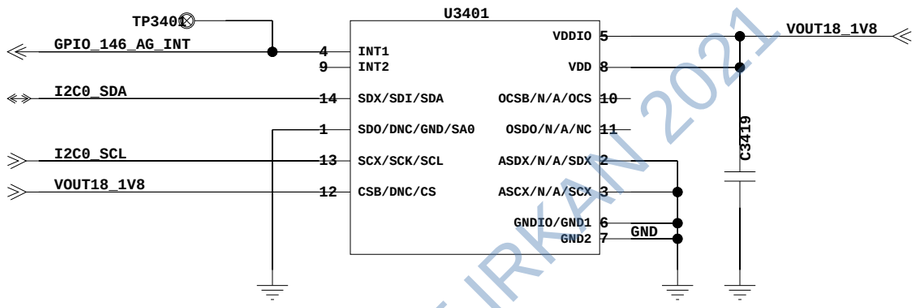
33. Smart_PA Reserved

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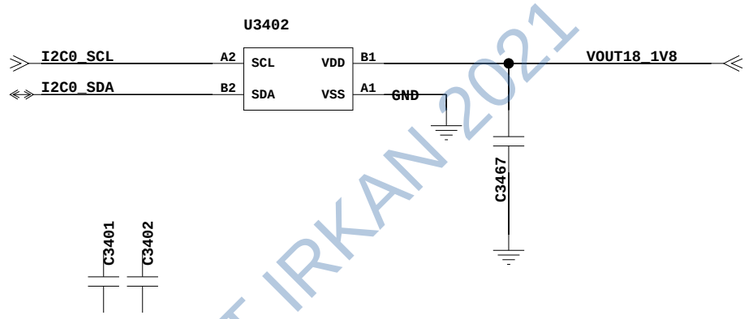
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34. X-Sensor

Accelerometer and Gyroscope 6-AXIS SENSOR



Compass

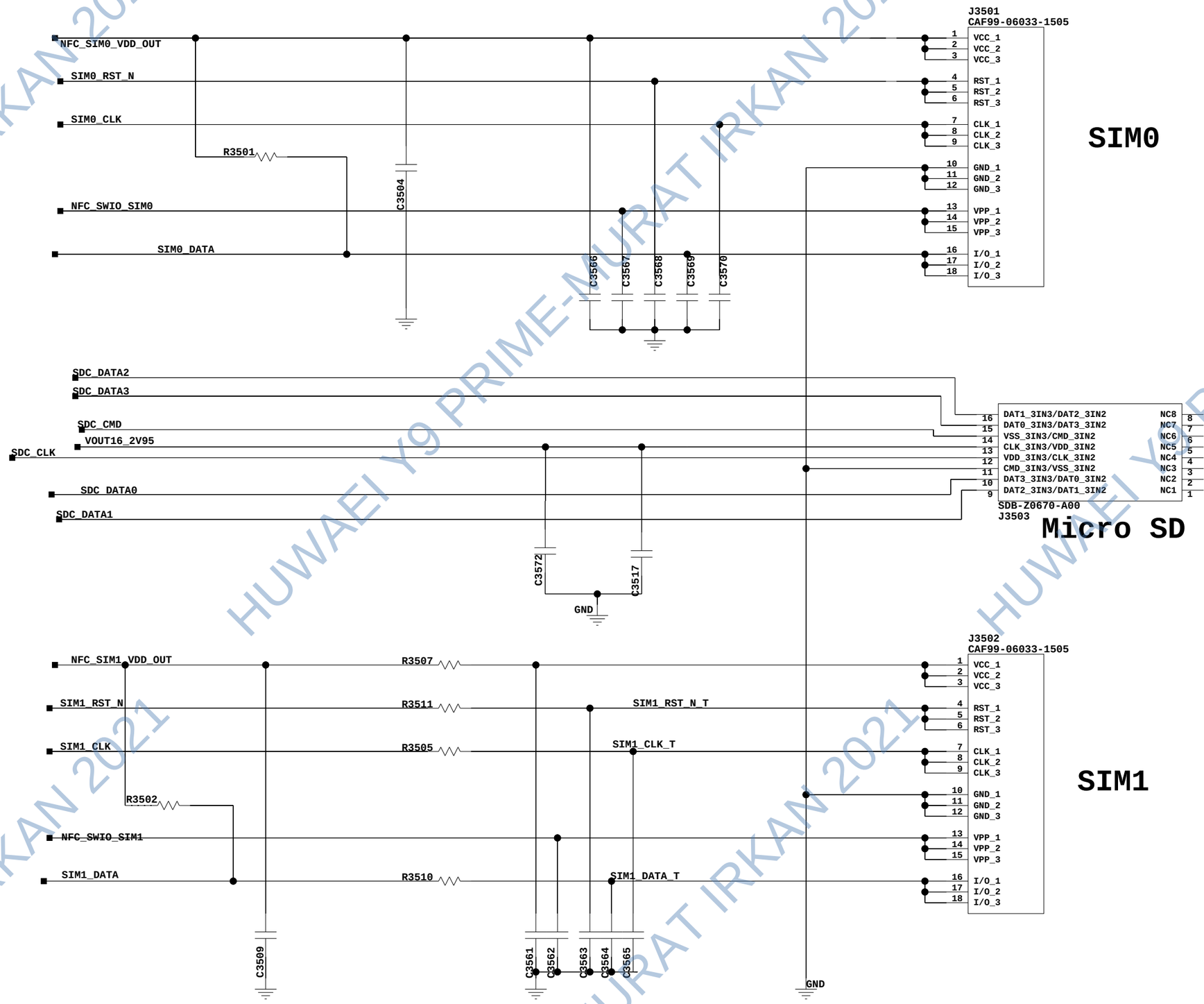


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35. SIM/uSD Card

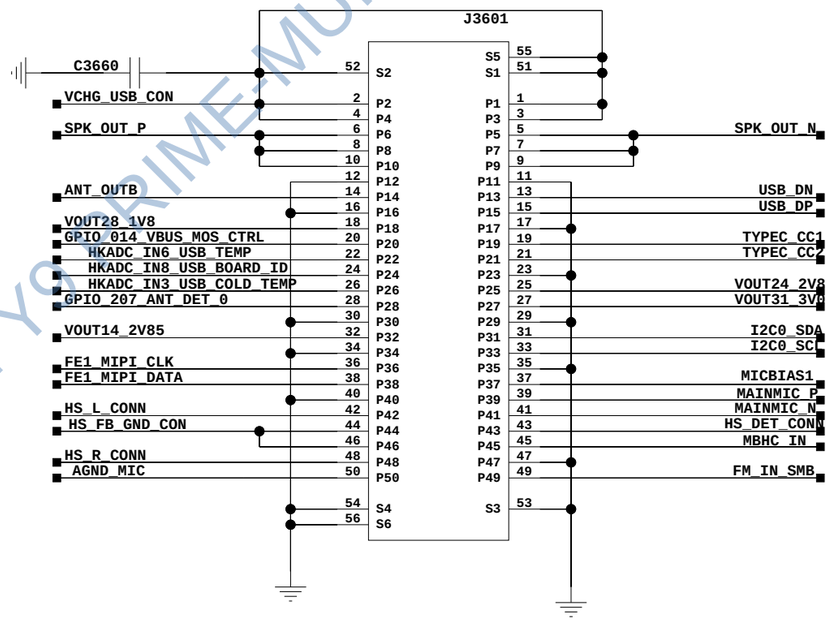


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36. FPC Interface 1

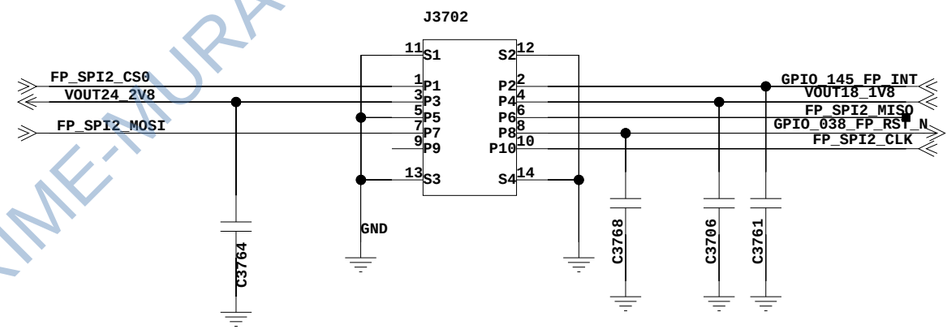
Main FPC BTB



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37. FPC Interface 2

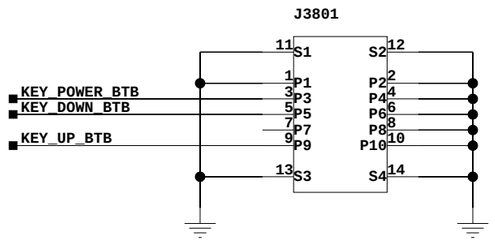
FingerPrint ZIF



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38. LED/Key

KEY BTB





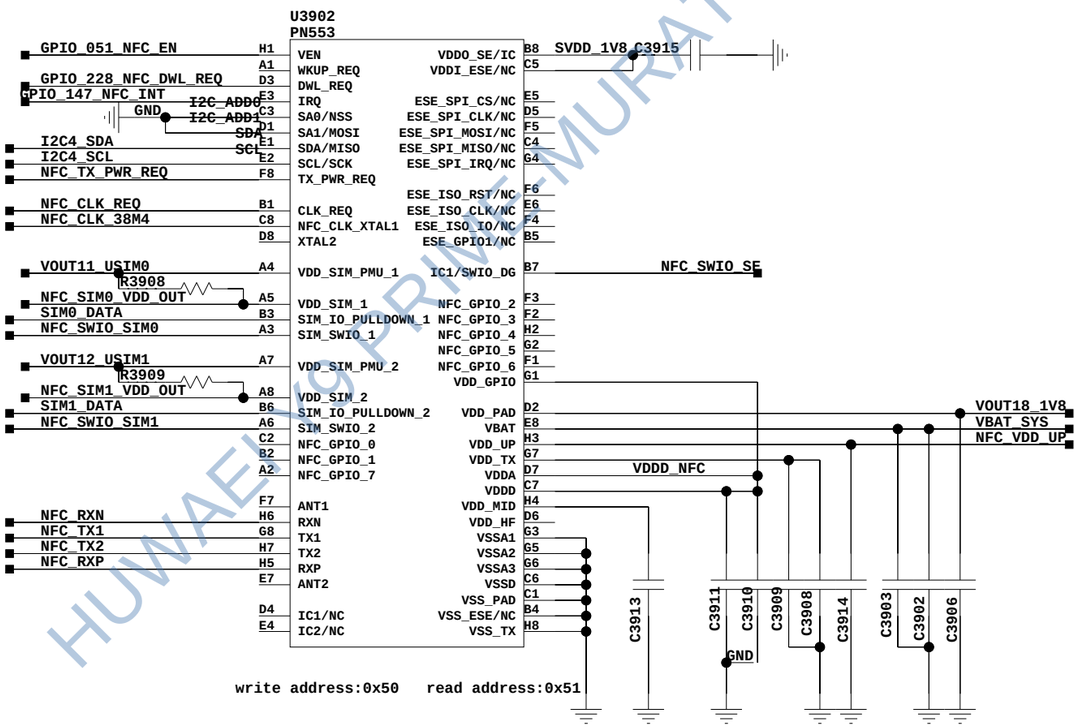
ｲ｢ﾐﾐｰﾚｷﾅ｣ｬﾓ　蒐鉾　ﾀｭｿｪｾ狢　ｬｷﾀﾖｹﾌ

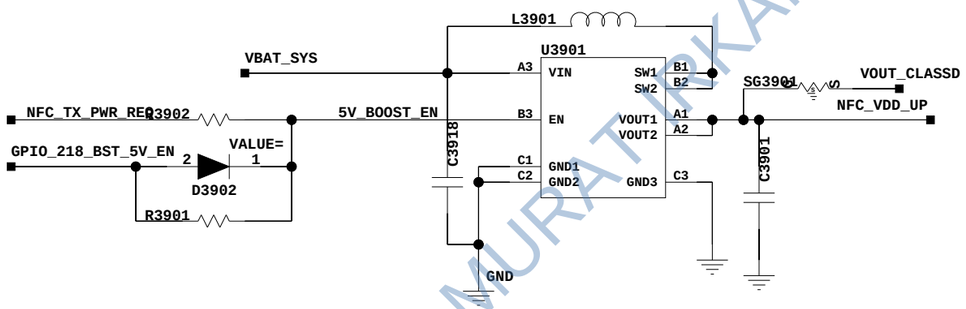
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39. NFC_BB

page		With NFC	No NFC
page12	R1250	DNI	SMT
page39	R3908	DNI	SMT
	R3909	DNI	SMT
	R3902	SMT	DNI
	D3902	SMT	DNI
	R3901	DNI	SMT
	C3902	SMT	DNI
	U3902	SMT	DNI
	C3903	SMT	DNI
	C3906,C3908	SMT	DNI
	C3909,C3910	SMT	DNI
	C3911	SMT	DNI
	C3913,C3914,C3915	SMT	DNI
	U3901	39111034	39110908



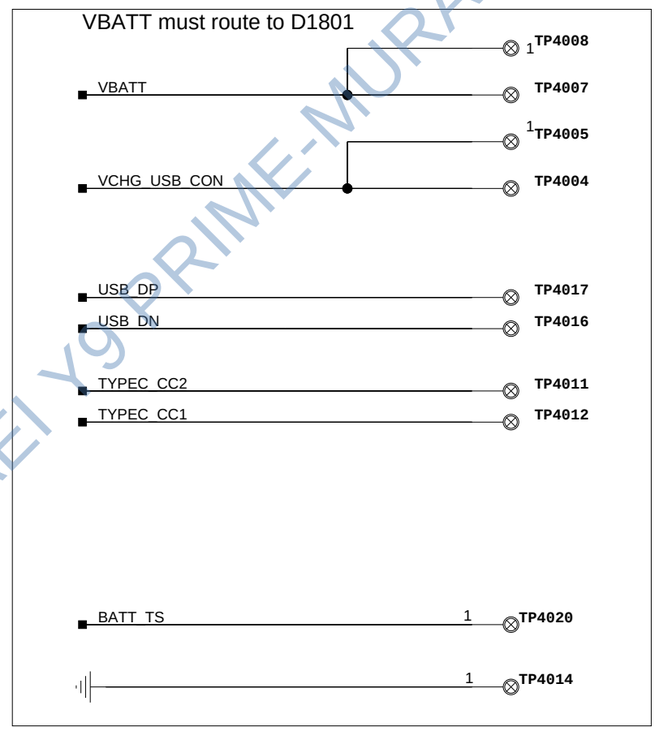


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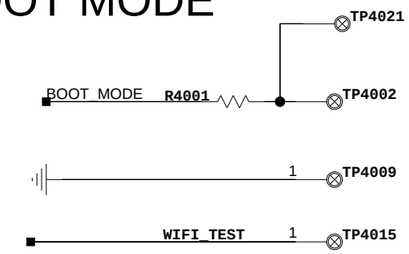
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40. Test Points/Shields

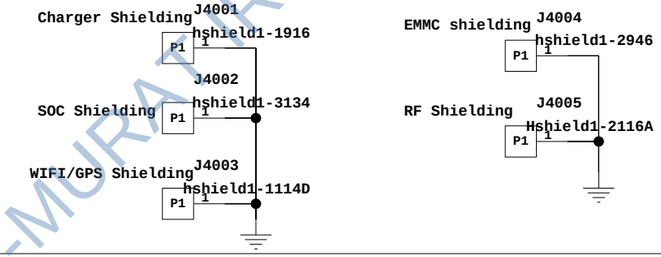
CBT/PT



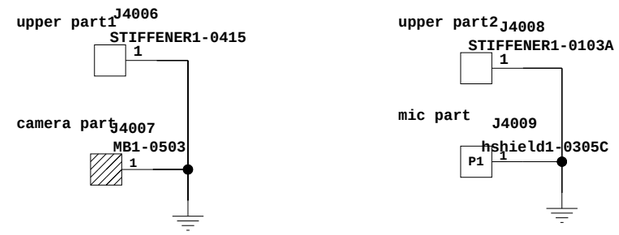
BOOT MODE



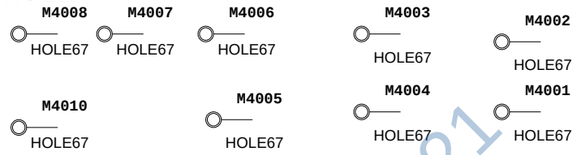
Shielding



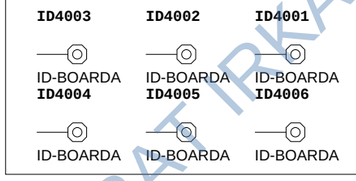
GND springs



HOLE



MARK POINT

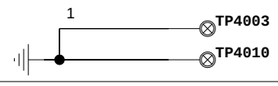


FUNCTION TEST

LCD BL

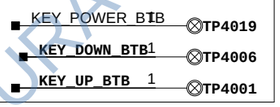
MICBIAS

GND



HEADPHONE

KEY



SPK

TYPE CC

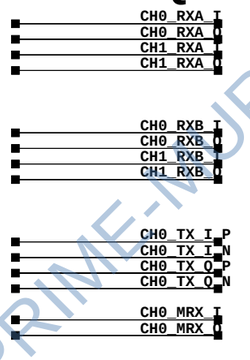
VIBRATOR

RCV

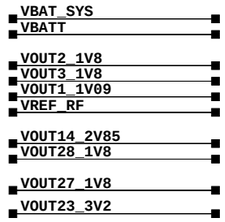
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41 Interface

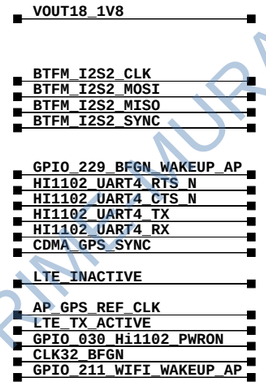
RFIC IQ



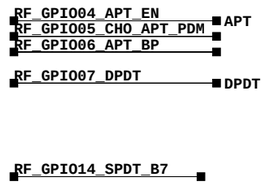
POWER



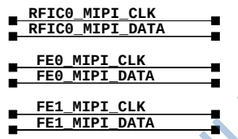
NC Interface



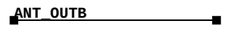
GPIO Interface



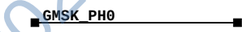
MIPI Interface



ANT Interface



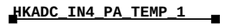
GSM



CLK

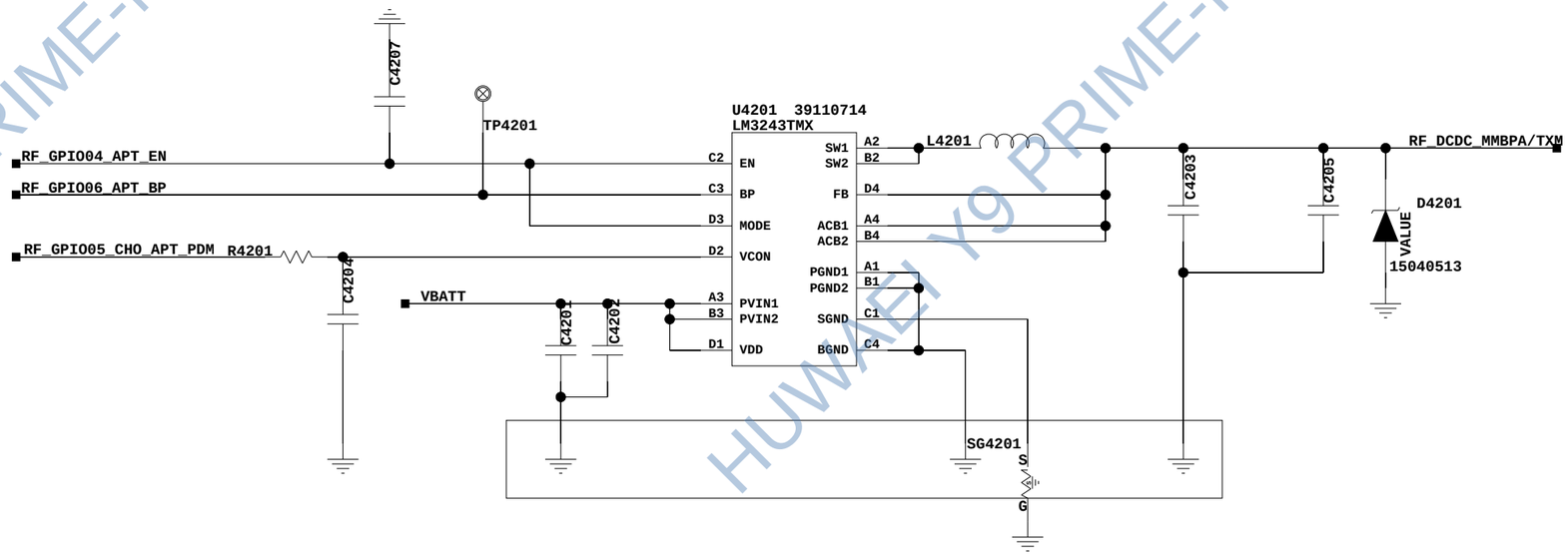


HKADC



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42 RF_PMU



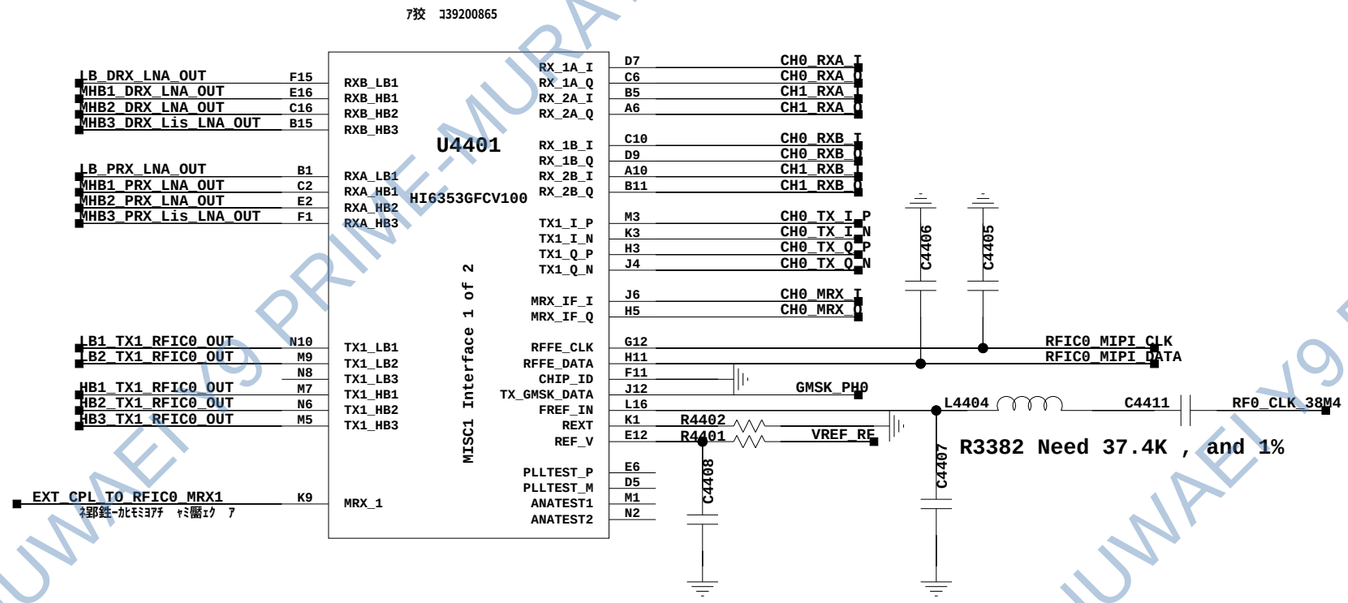
1.DCDCｵ酉丁ｷﾎｪPAｷﾀｻ、ｹ鯣ｻｻｯｵ酉丁ｷｵﾄﾗ魑ﾉｲｿｷﾖ
2.ｲｼﾏﾟﾊｱﾏﾈｾｭｹ　ﾎﾈﾑｹｹﾜD4201」ｬﾔﾙﾐﾇﾐﾍﾗﾟﾏﾟｸ　AｺﾍTXMｹｳｵ
3.ｲｼﾏﾟﾊｱｿｼﾂﾇﾏﾟｿ恢ｬDCDCｸ　SMｹｳｵ醋ｱ」ｬﾐ睢」ﾖ、ﾍｲﾁ　　ｽ3A
4.ﾊ莎　酳ﾝﾓ　ﾈﾑｹｹﾜｵﾄGNDﾓ　昀羞ﾘｸ　　ｬﾖｱｽﾓﾏﾂﾖ　ﾘ
5.ｹｳｵ釋ﾟﾏﾟﾇｿｸﾉﾈﾅﾇﾒｻ盒　　Aﾐｳｲｨ」ｬｽｲﾒ麹魄　ﾘｴﾇﾀ

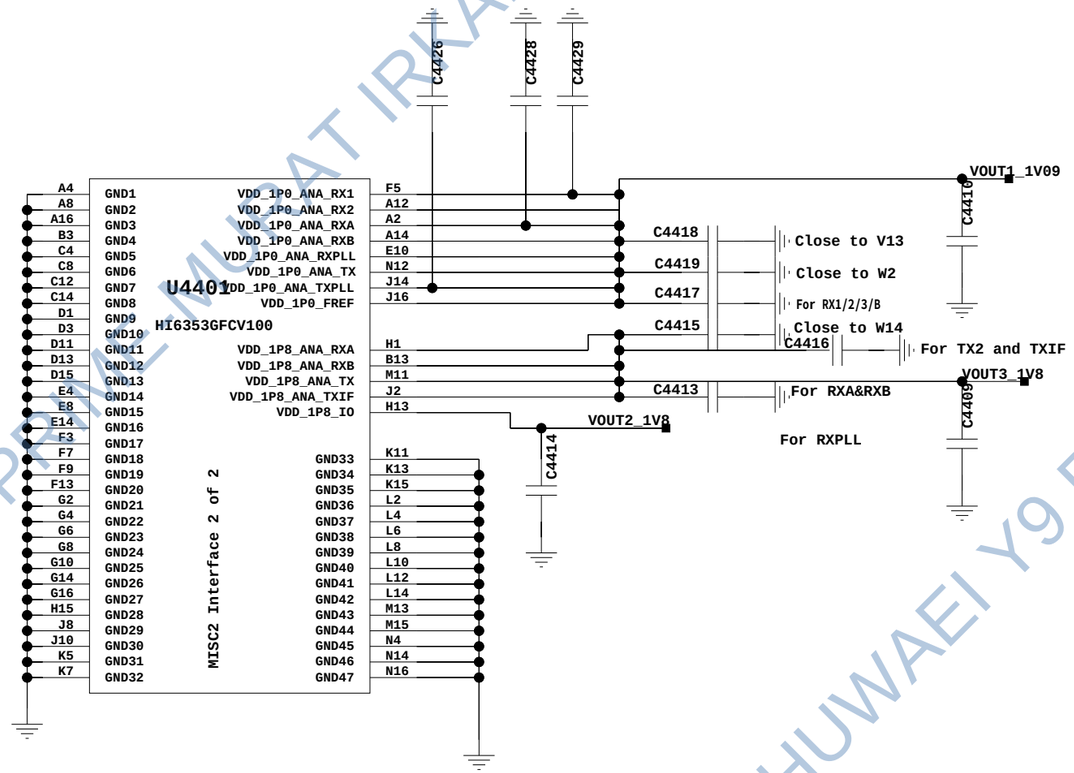
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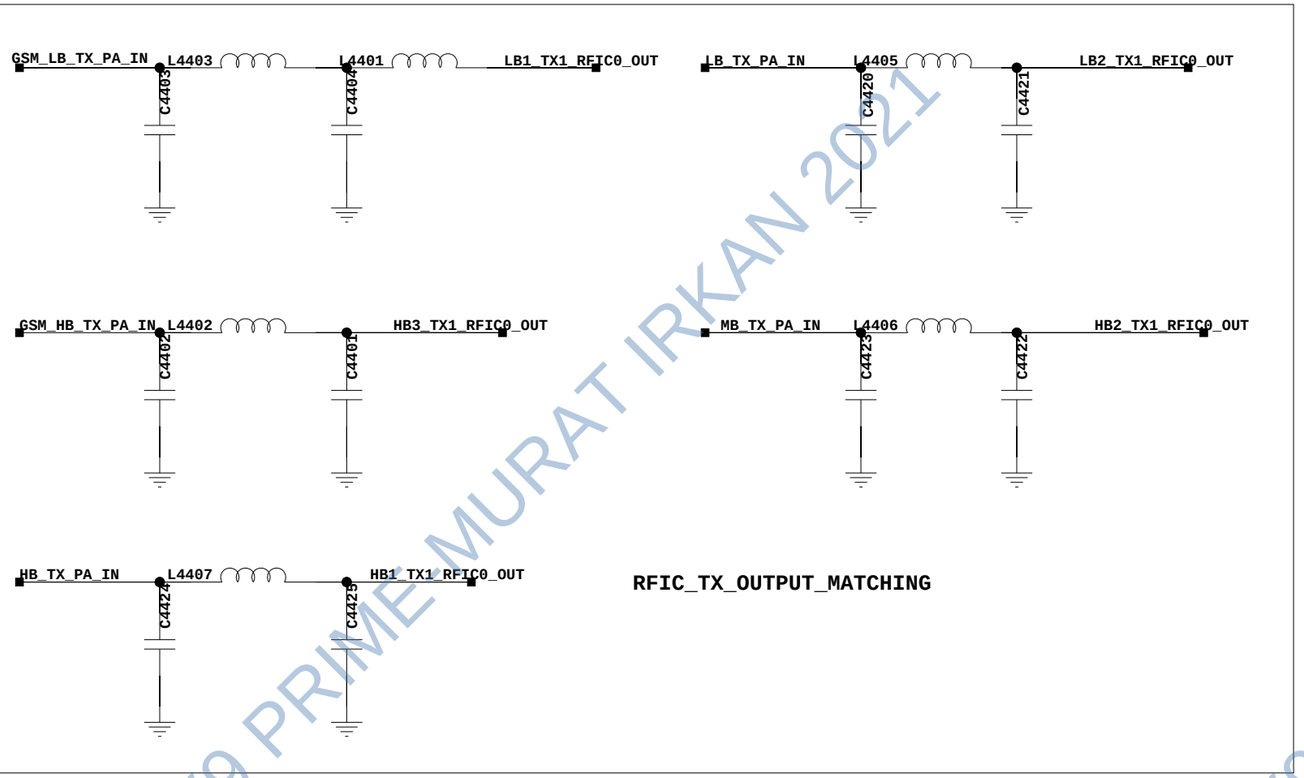
43 RF_PMU_Reserved

HUWAEI Y9 PRIME-MURAT IRKAN 2021
HUWAEI Y9 PRIME-MURAT IRKAN 2021
HUWAEI Y9 PRIME-MURAT IRKAN 2021
HUWAEI Y9 PRIME-MURAT IRKAN 2021
Mobile1Tech.com

44.RFIC0_Hi6353





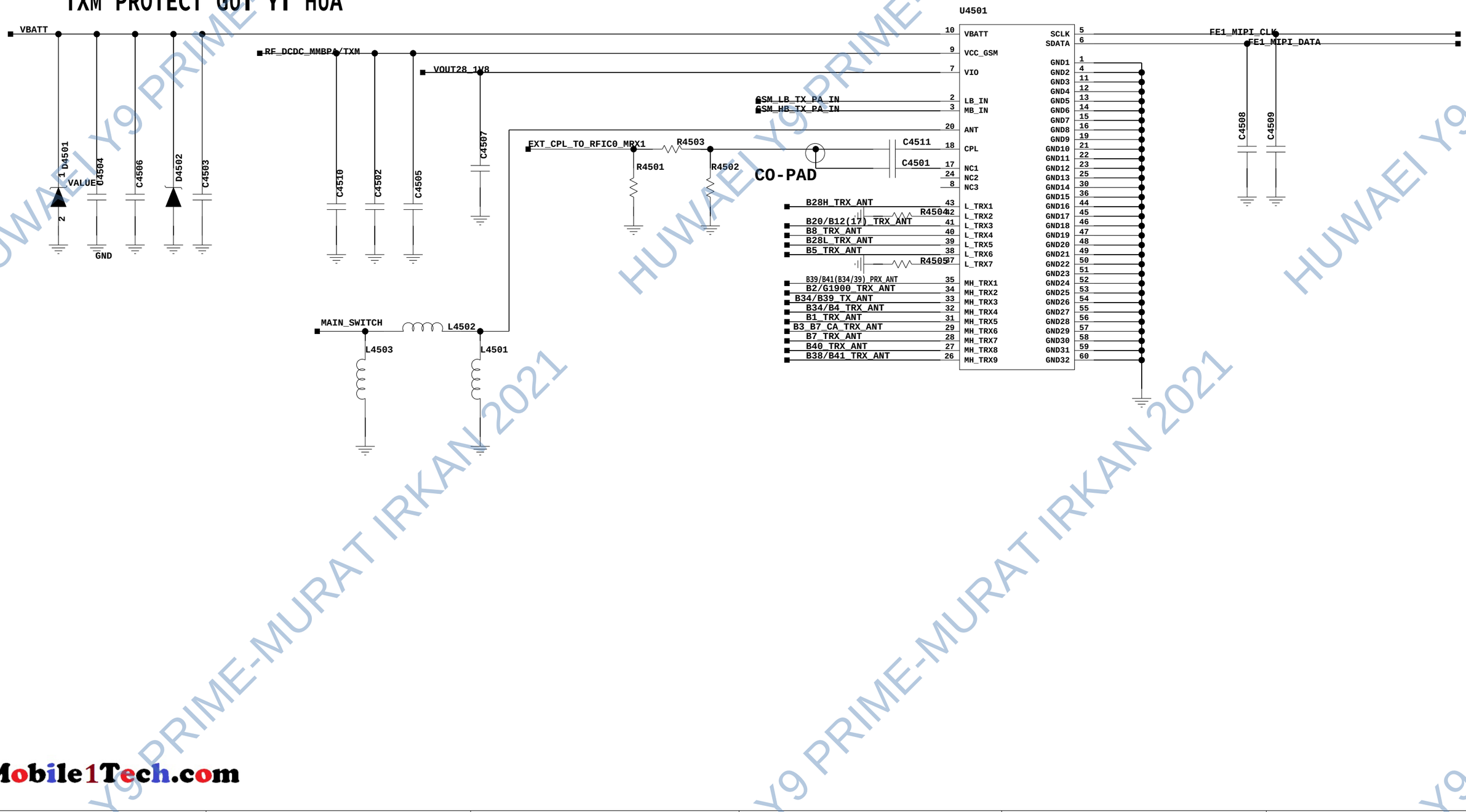


HUWAEI Y9 PRIME-MURAT IRKAN 2021
Mobile1Tech.com

45 PA0_TXM

TXM PROTECT GUI YI HUA

HI6D21 47150693

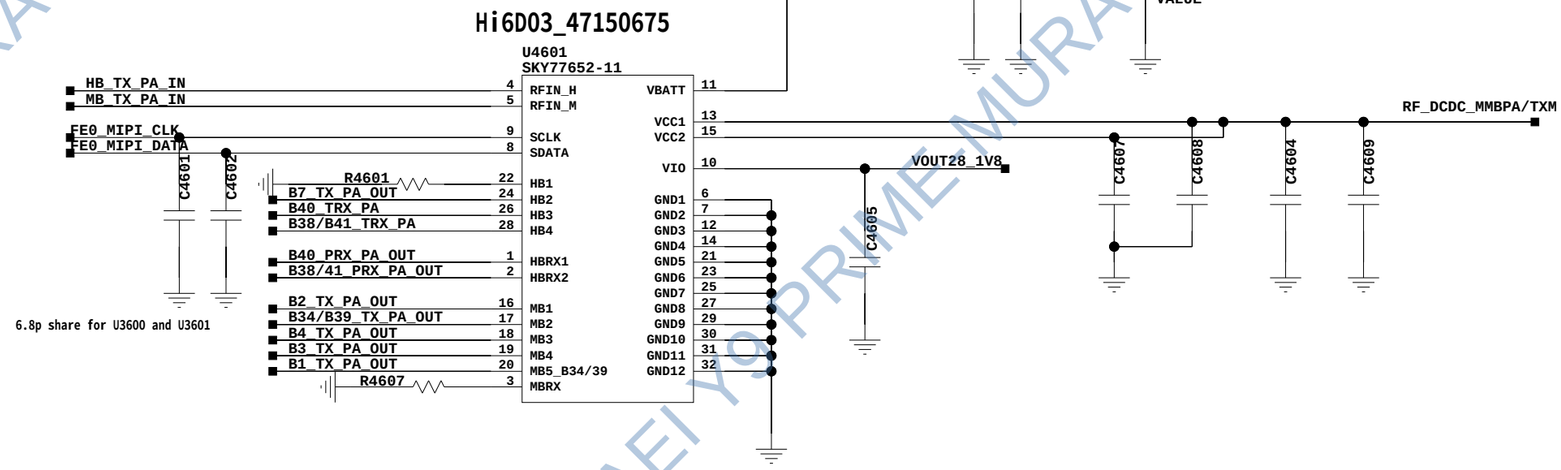


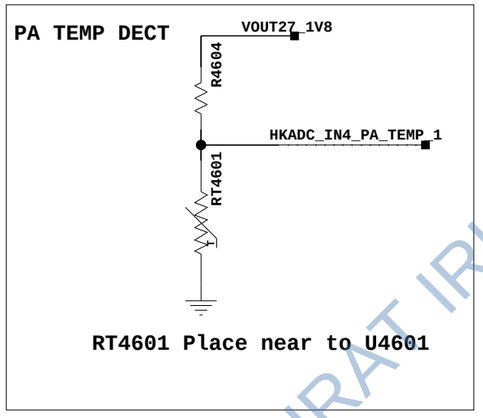
Mobile1Tech.com

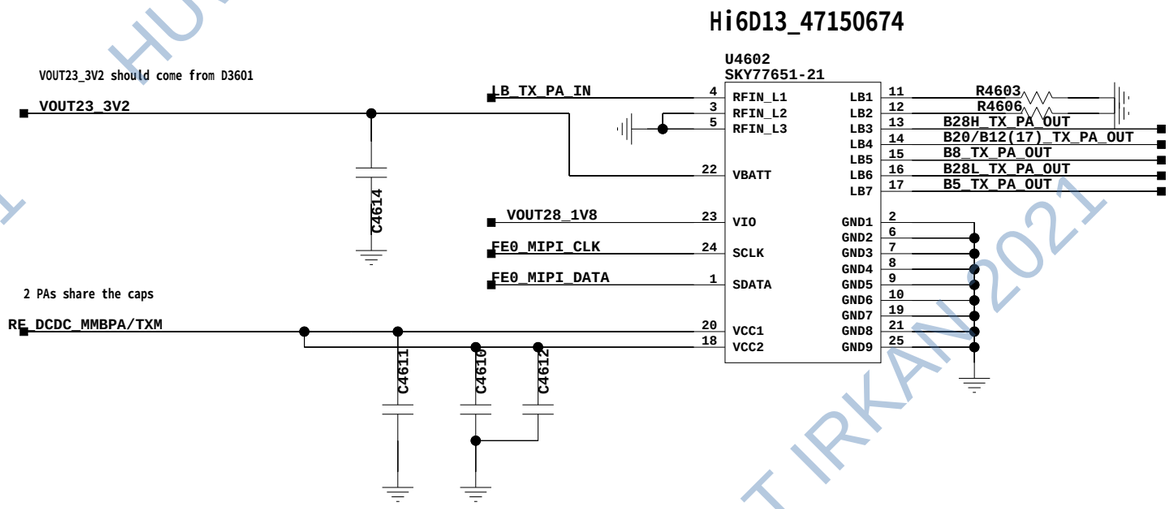
HUWAEI Y9 PRIME-MURAT IRKAN 2021

46 PA1_MMBPA

MMMBPA protect gui yi hua

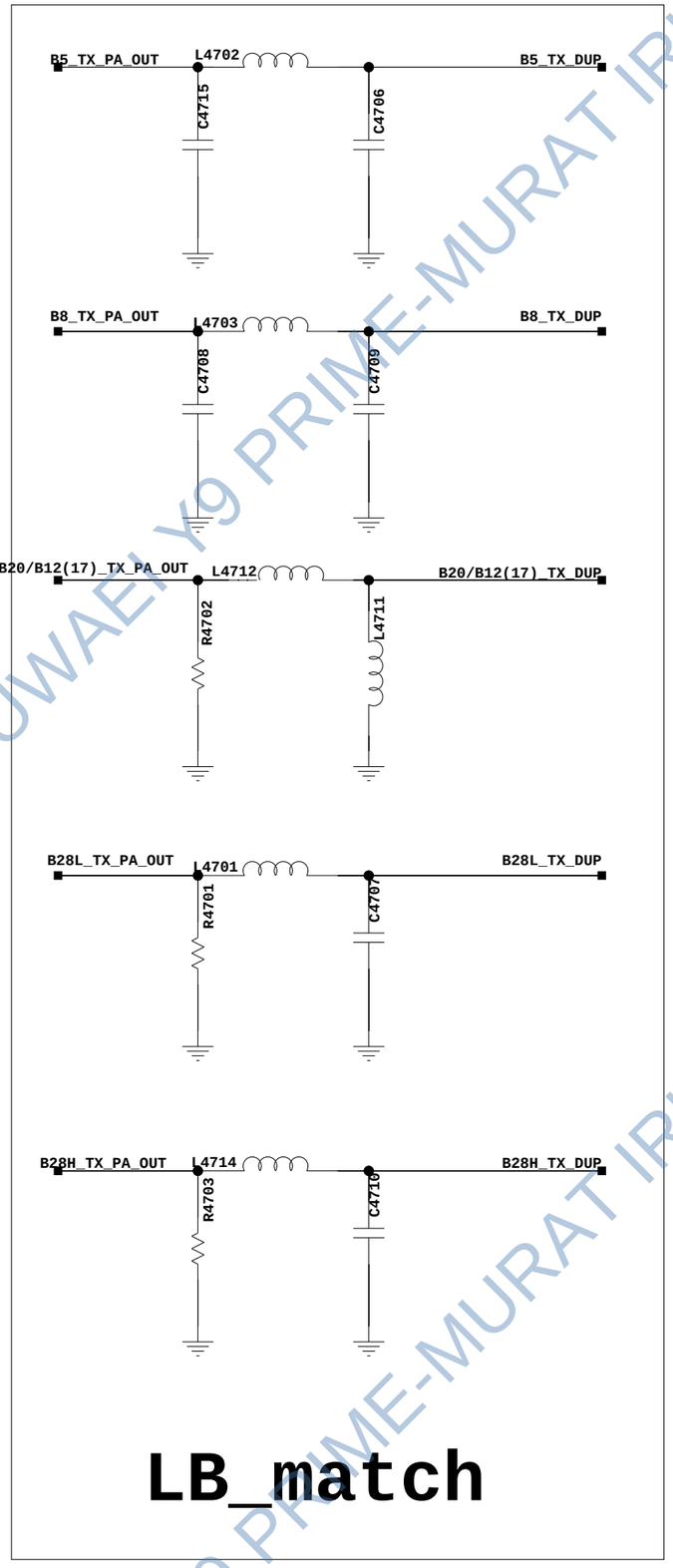


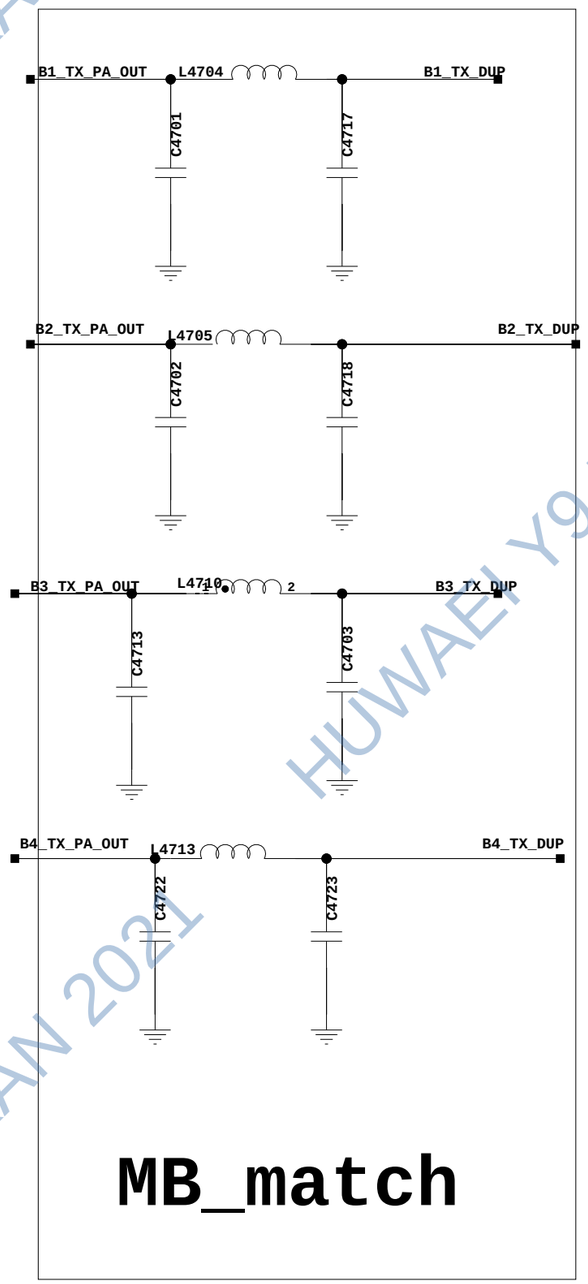


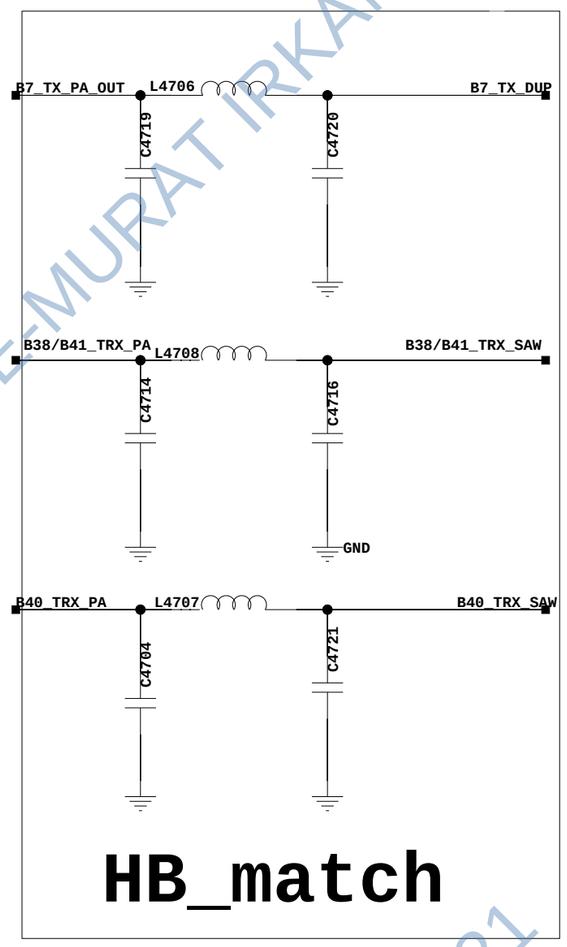


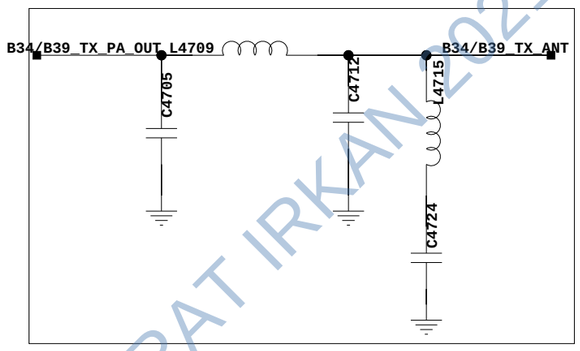
HUWAEI Y9 PRIME-MURAT IRKAN 2021
Mobile1Tech.com

47 PA2_TX_Loadpull









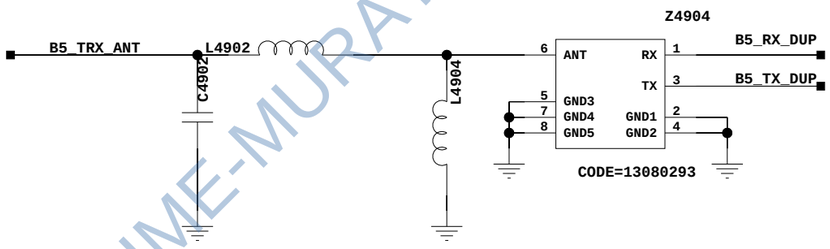
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Mobile1Tech.com

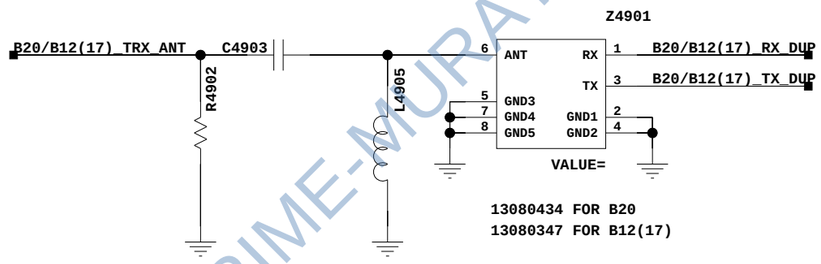
48 PA3_Reserved

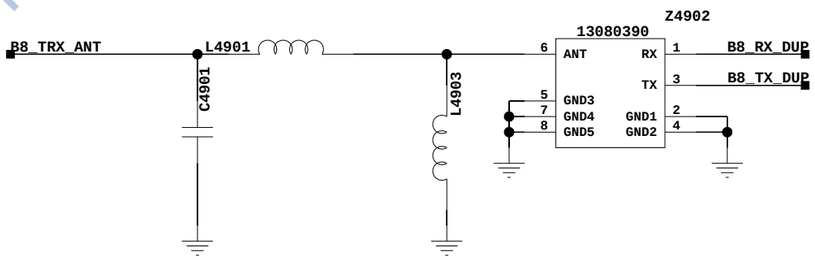
HUWAEI Y9 PRIME-MURAT IRKAN 2021

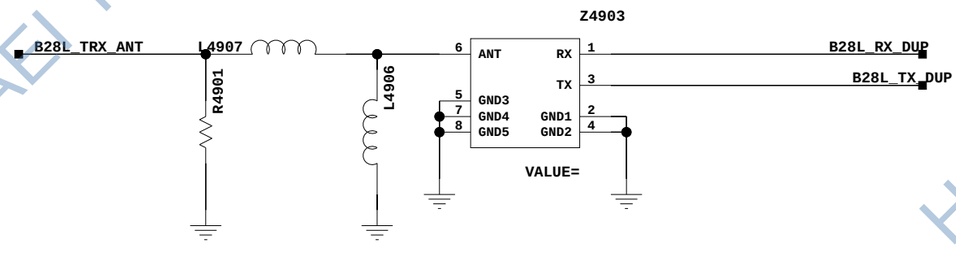
Mobile1Tech.com

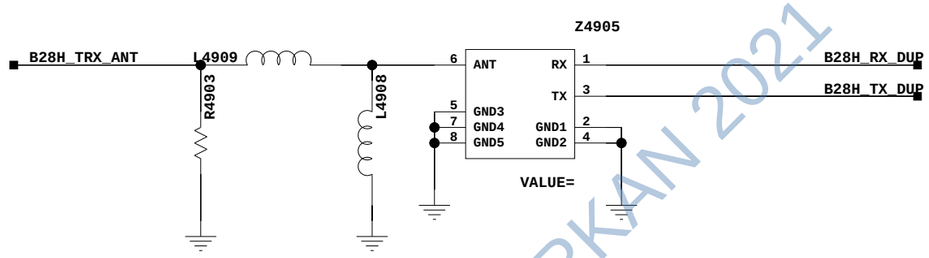
49 TRX_LB_1











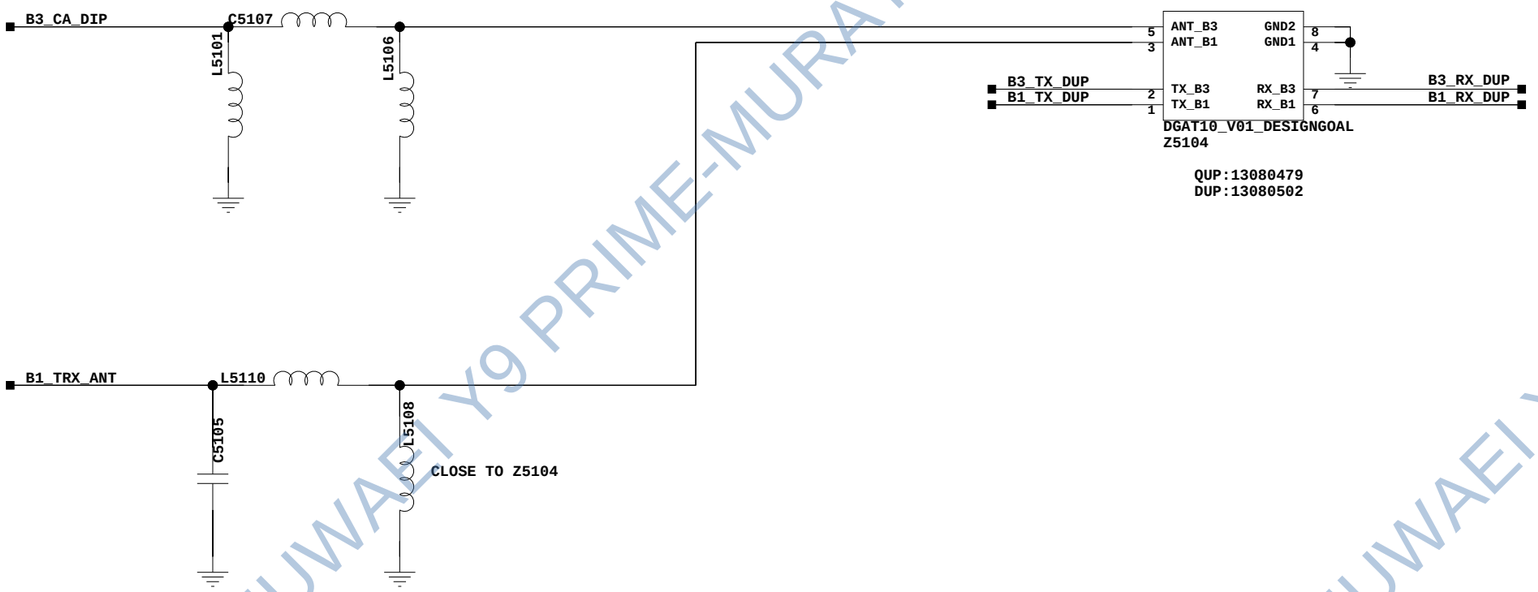
HUWAEI Y9 PRIME-MURAT IRKAN 2021
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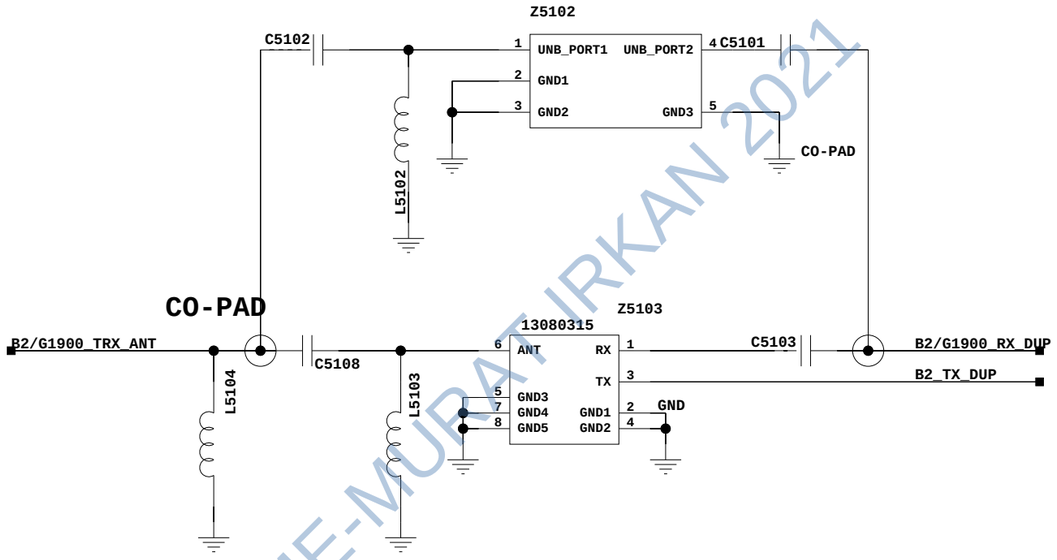
50 TRX_LB_2_Reserved

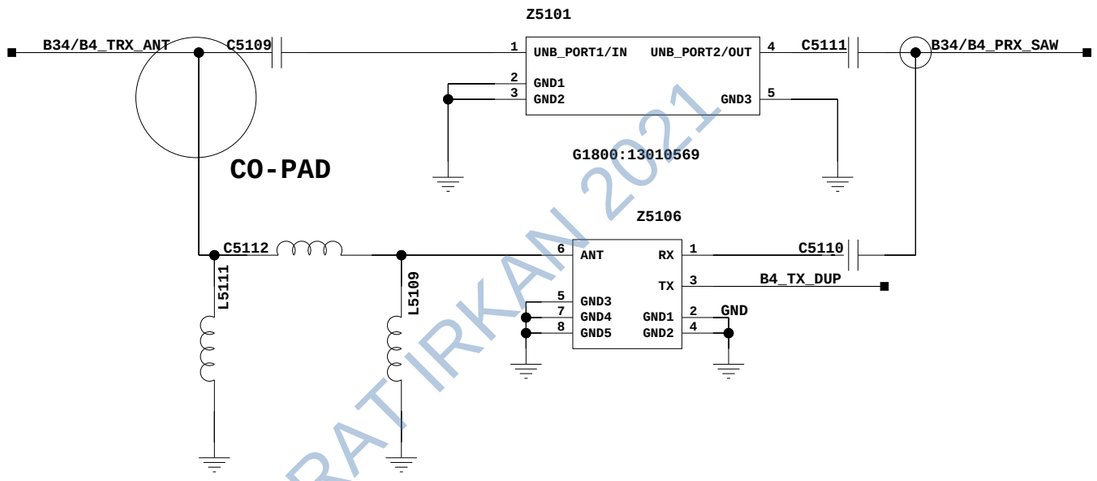
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51 TRX_MB_1







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52 TRX_MB_2_Reserved

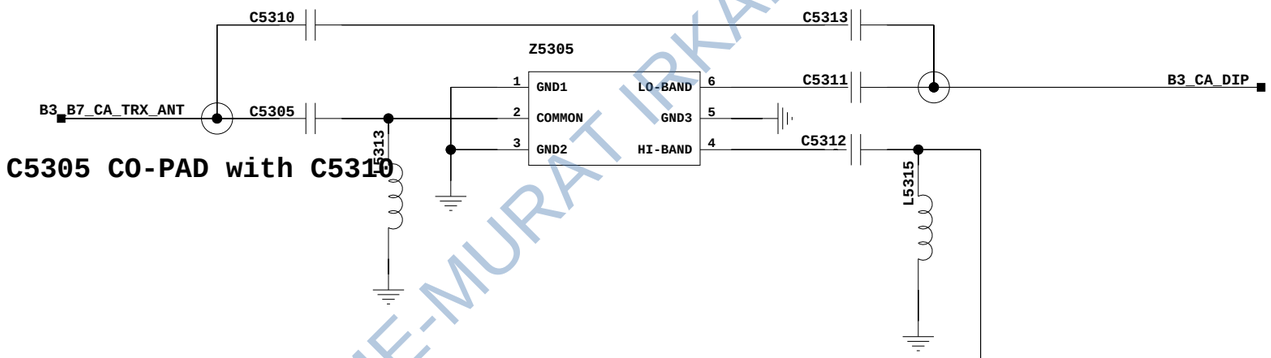
HUWAEI Y9 PRIME-MURAT IRKAN 2021

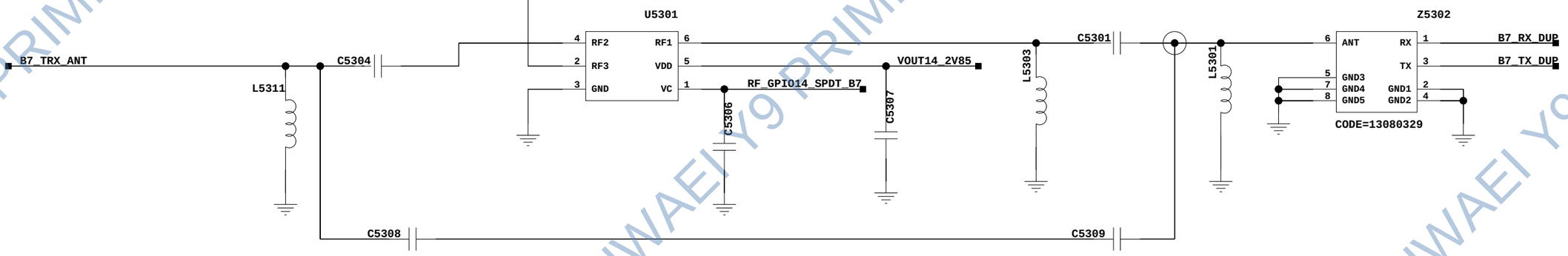
Mobile1Tech.com

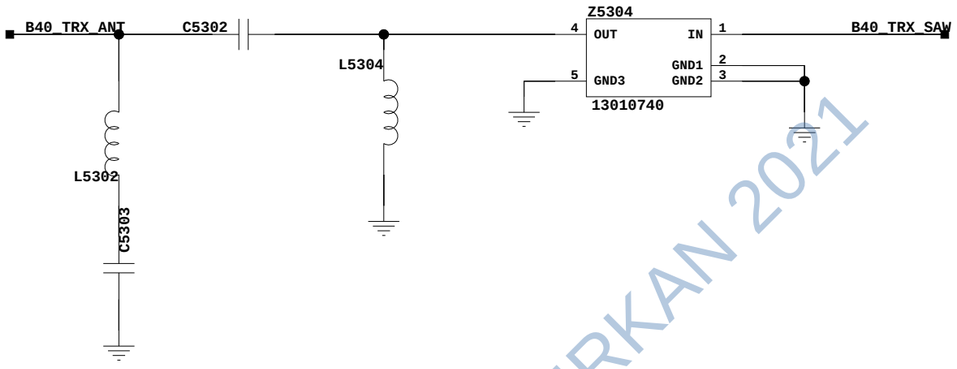
53 TRX_HB_1

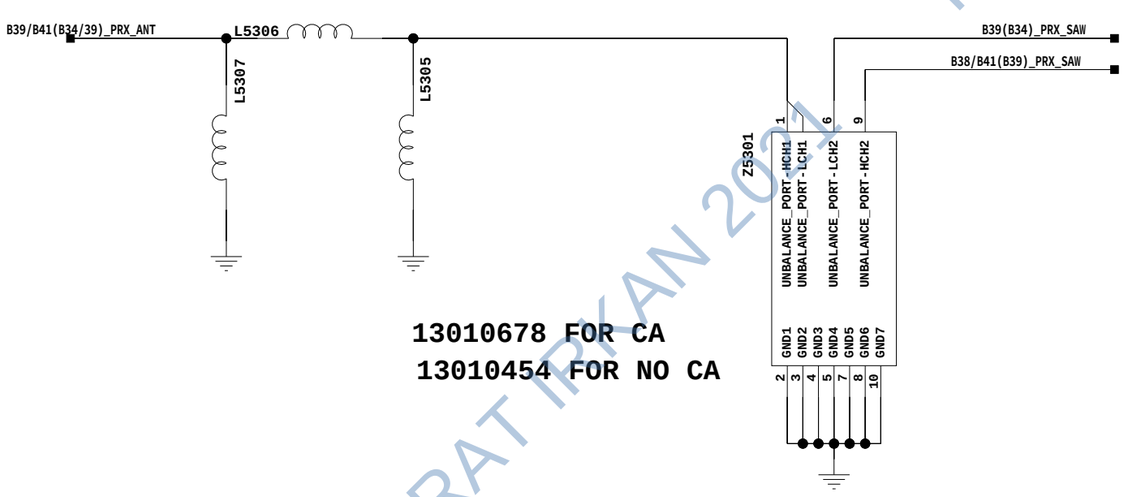
C5513 CO-PAD with C5311

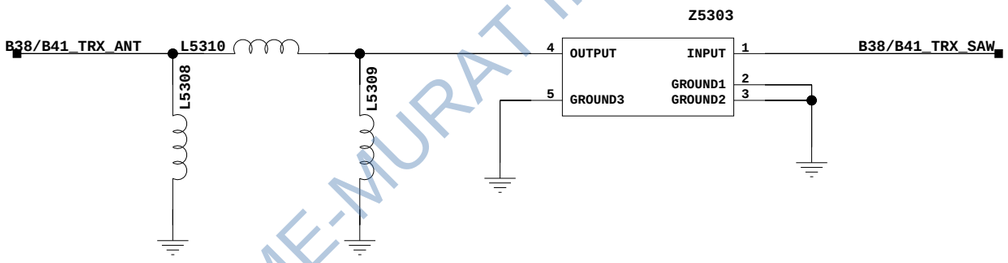
C5305 CO-PAD with C5310











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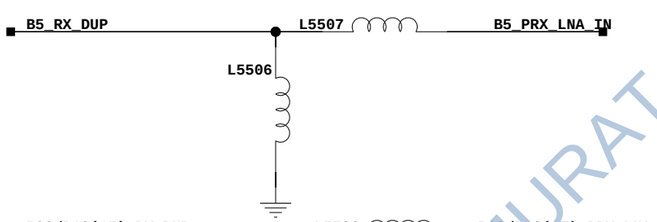
Mobile1Tech.com

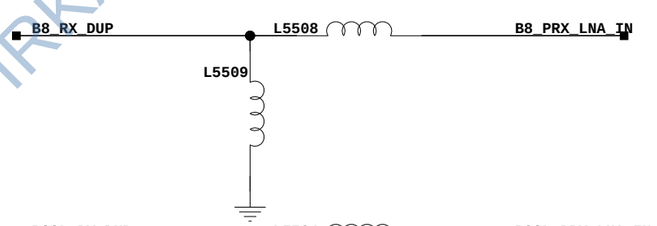
54 TRX_HB_2_Reserved

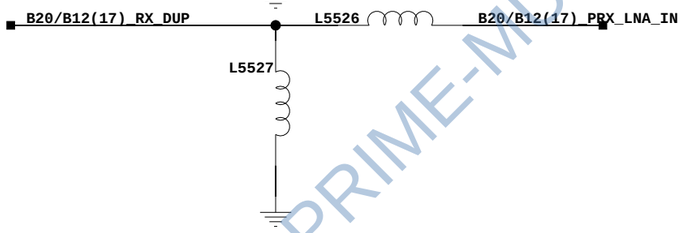
HUWAEI Y9 PRIME-MURAT IRKAN 2021

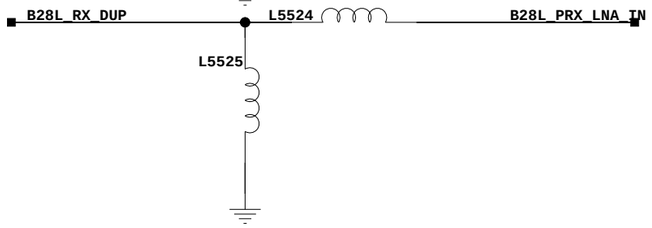
Mobile1Tech.com

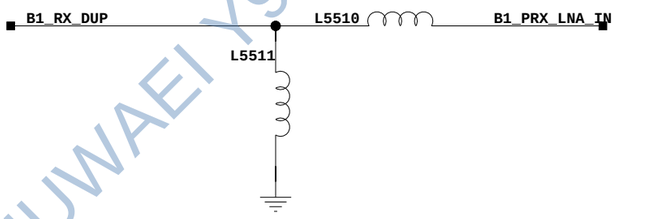
55 PRX_LNA_Module

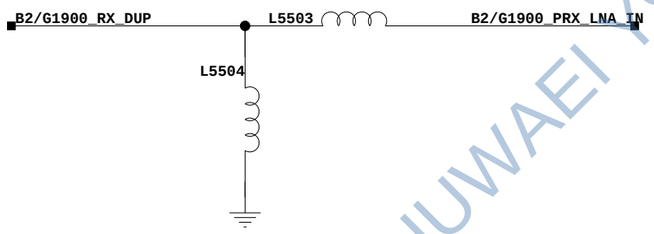


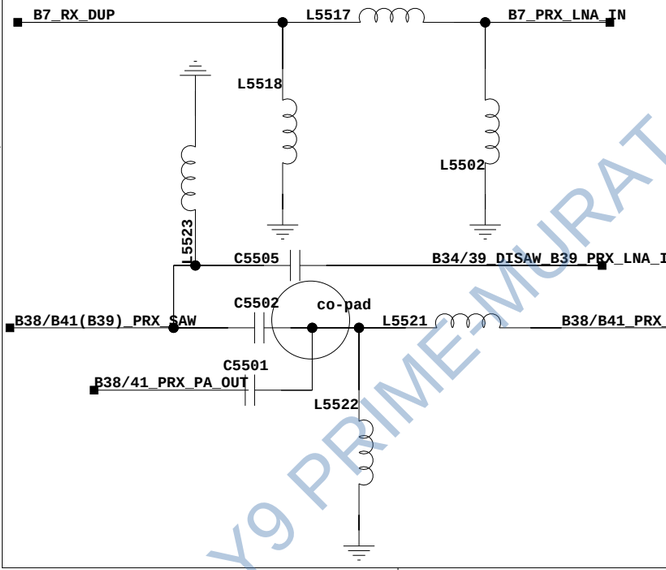


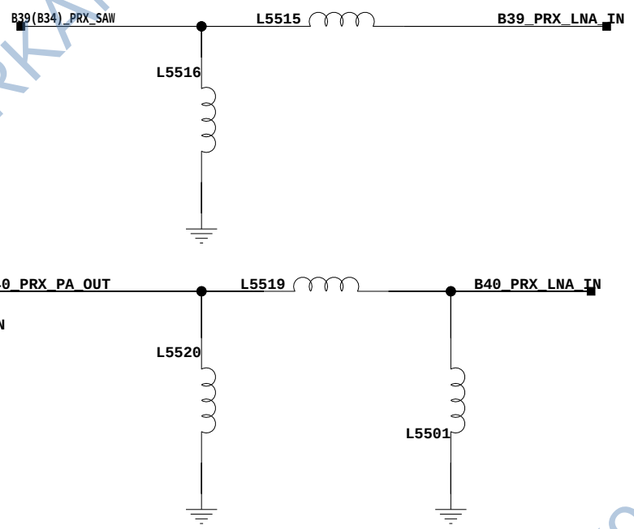


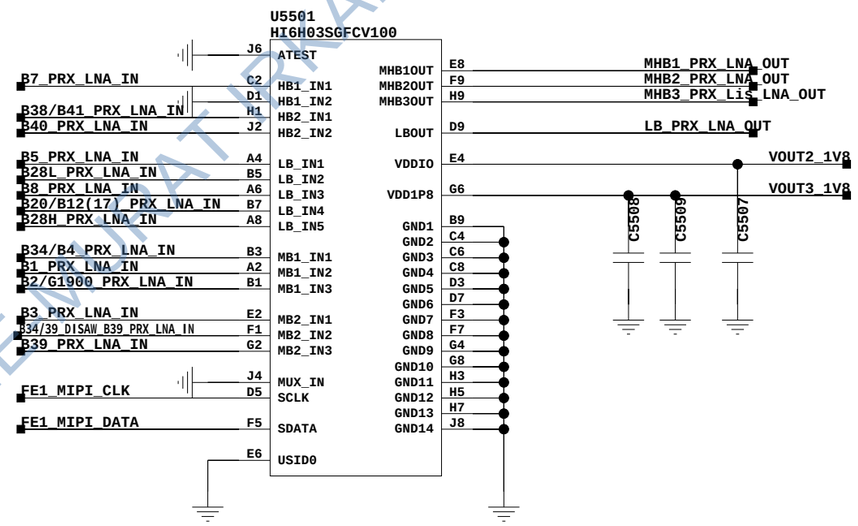




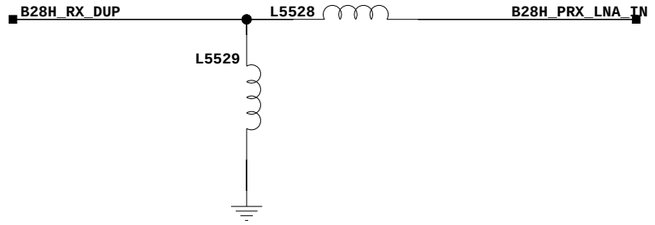








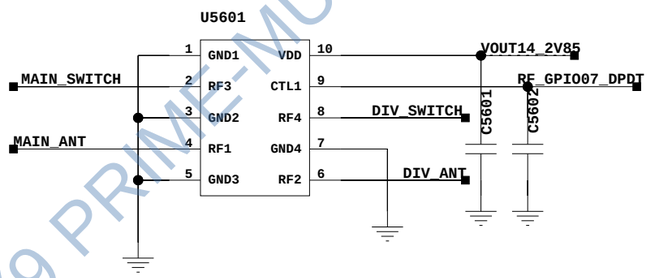
The Inputport unused should be connected to GND!

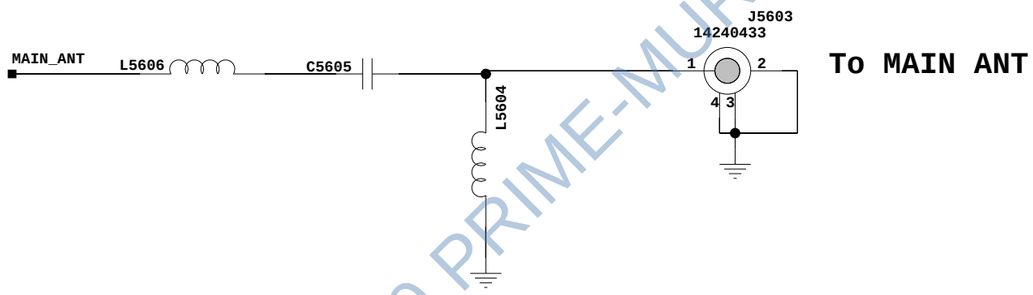


6H03 Frequency Range	
LB_IN1	B5
LB_IN2	703-894MHz
LB_IN3	B8
LB_IN4/5	717-821MHz
MB1_IN1	1805-2200MHz B40
MB1_IN2/3£¬MB2_IN1/3	1805-2200MHz
HB1/2_IN1	B41/38/7
HB1/2_IN2	2300-2690MHz

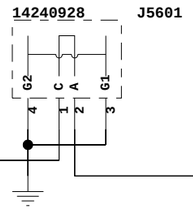
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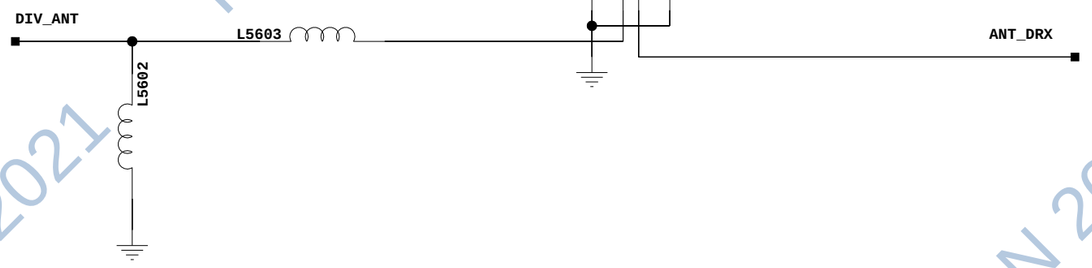
56 MAIN_DPDT





SKY13699-21 Truth Table	
CTRL1	State
0	RF1-RF3,RF2-RF4
1	RF1-RF4,RF2-RF3





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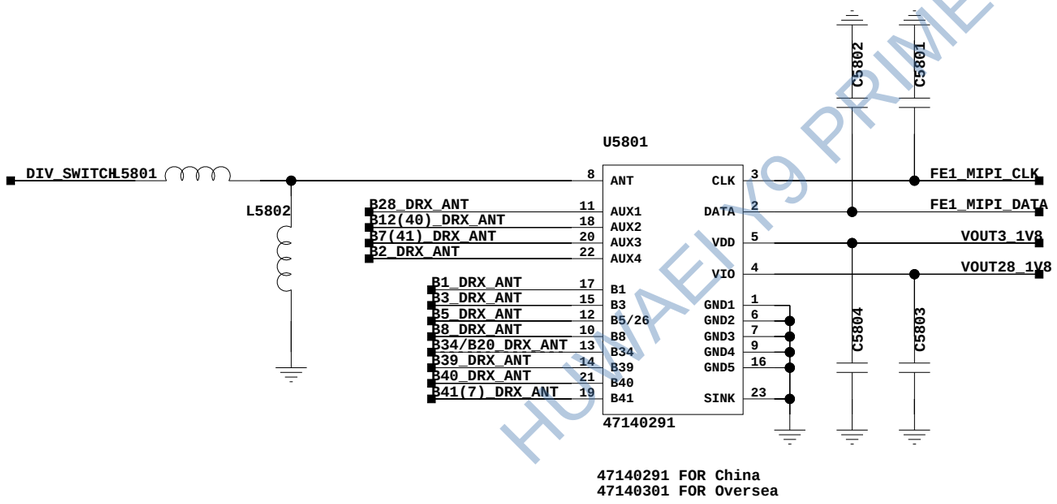
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57 MAIN_DPDT_Reseverd

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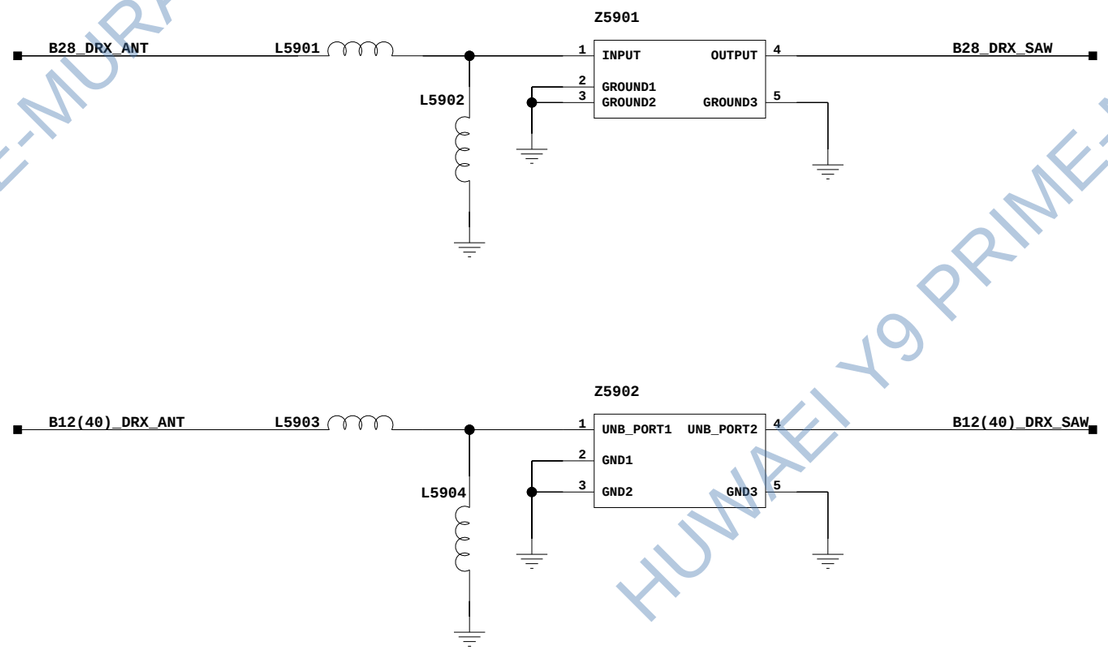
58 DRX_Switch



47140291 FOR China
47140301 FOR Oversea

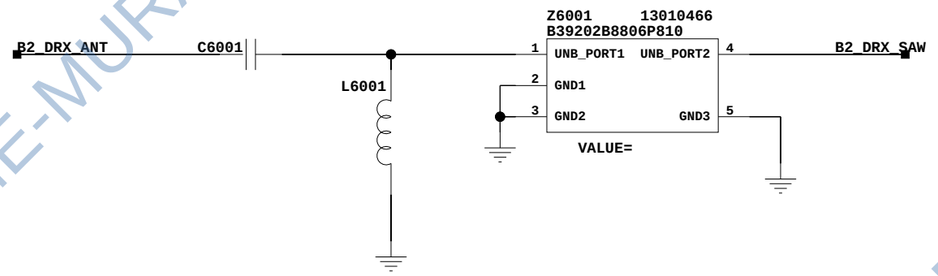
HUWAEI Y9 PRIME-MURAT IRKAN 2021
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59 DRX_LB



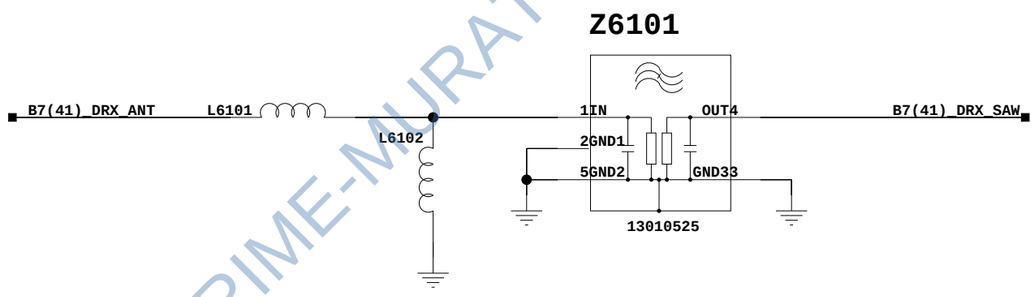
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60 DRX_MB



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61 DRX_HB



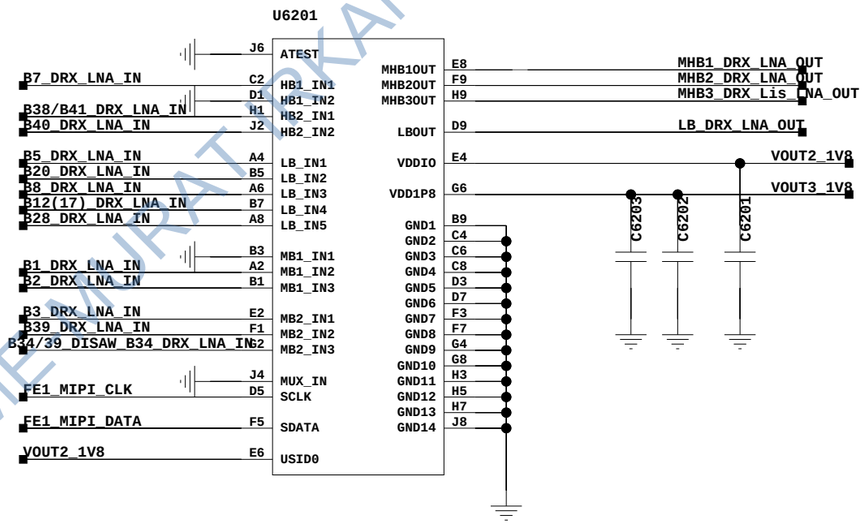
13010525 FOR B7
13010677 FOR B41

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62 DRX_LNA_Module



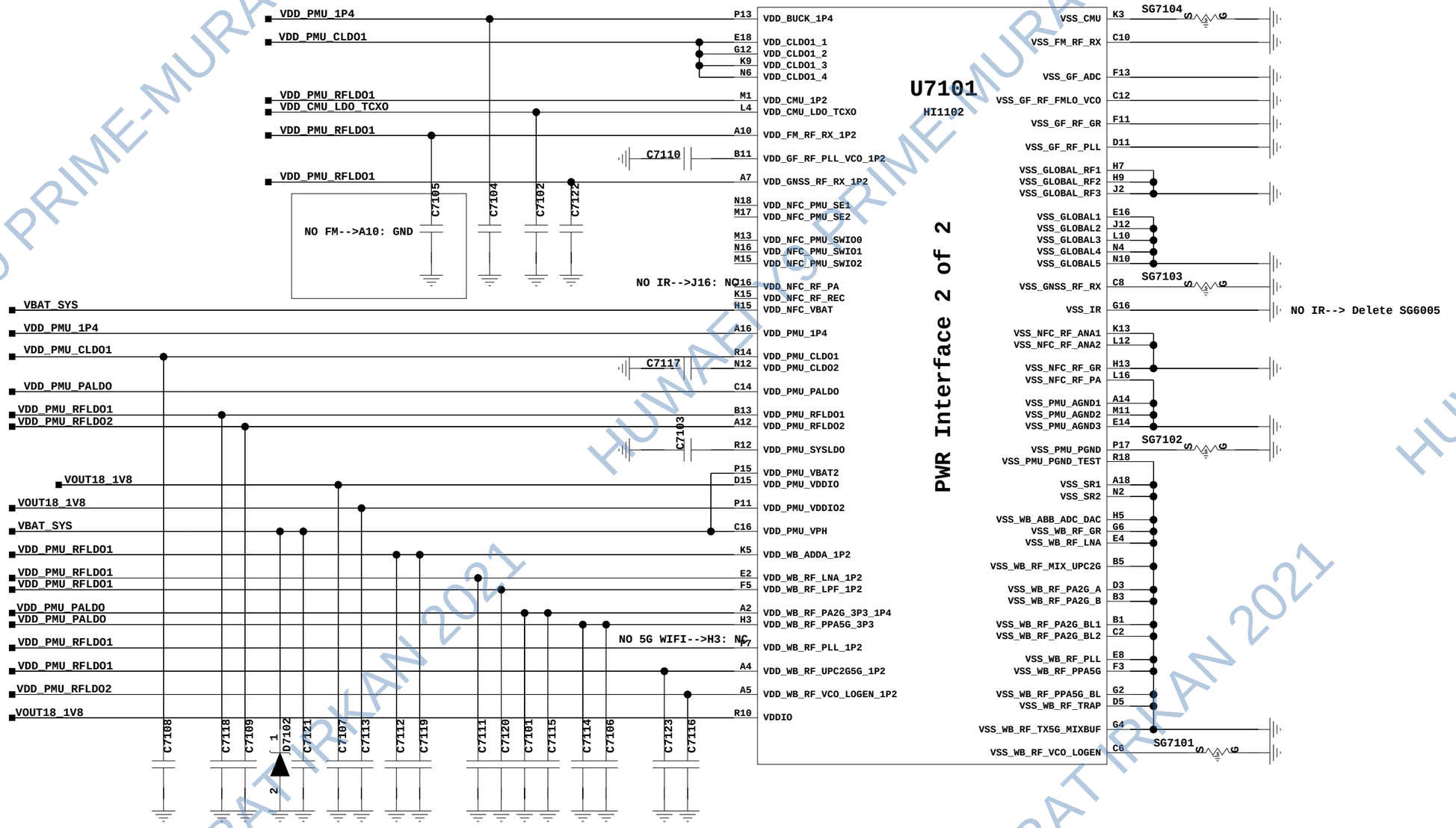
The Inputport unused should be connected to GND!



6H03 Frequency Range	
LB_IN1	B5
LB_IN2	703-894MHz
LB_IN3	B8
LB_IN4/5	717-821MHz
MB1_IN1	1805-2200MHz B40
MB1_IN2/3£¬MB2_IN1/3	1805-2200MHz
HB1/2_IN1	B41/38/7
HB1/2_IN2	2300-2690MHz

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71 NC_POWER



NO 5G-->Delete D6002

Add by Hisilicon for debug

VOUT18_1V8

VBAT_SYS

Boston,Miami: VOUT2->VOUT18_1V8

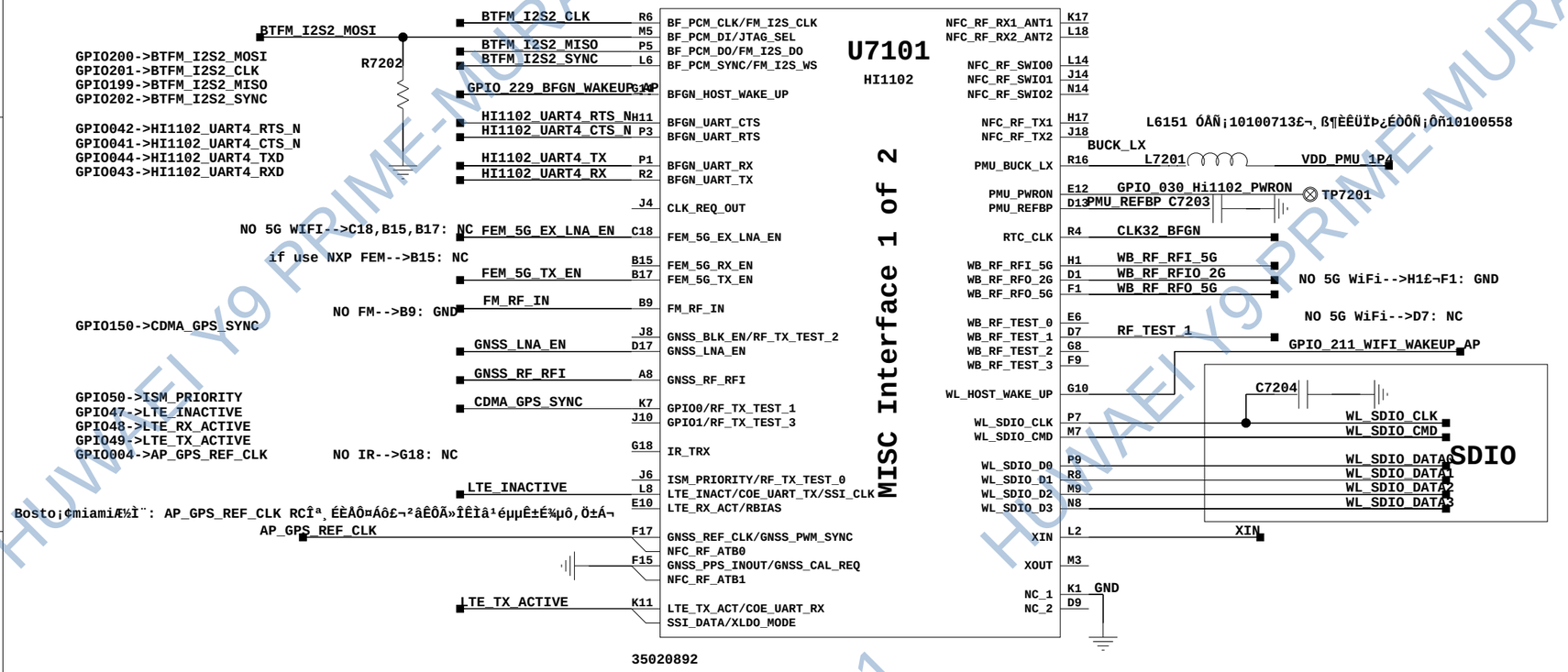
Chicago,Dallas: VOUT2

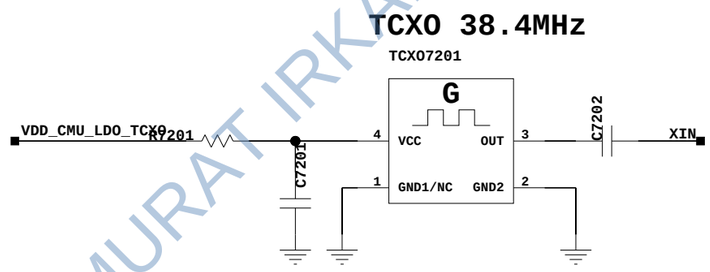
R6082¡¢R6083:Æ½Ì¨Ê×ÏîÄ¿ÊÔÖÆ±£ÁôÊ¬×îÕÕ¹éµµÊ±É¾µô,Ö±Á¬£»·ÇÆ½Ì¨Ê×ÏîÄ¿Ö±Á¬

HUWAEI Y9 PRIME-MURAT IRKAN 2021

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72 NC_BB_Miami



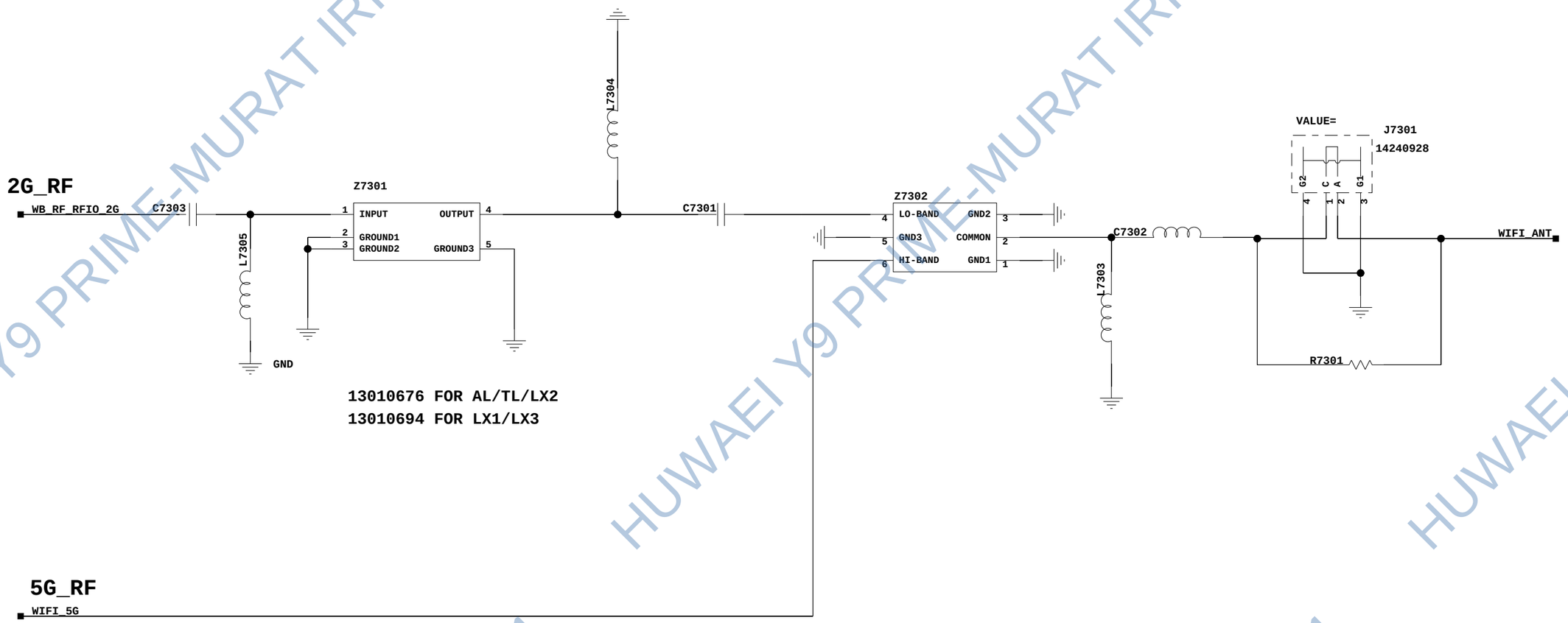


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بمناسبة إعادة فتح جامع آيا صوفيا الكبير 24/7/2020

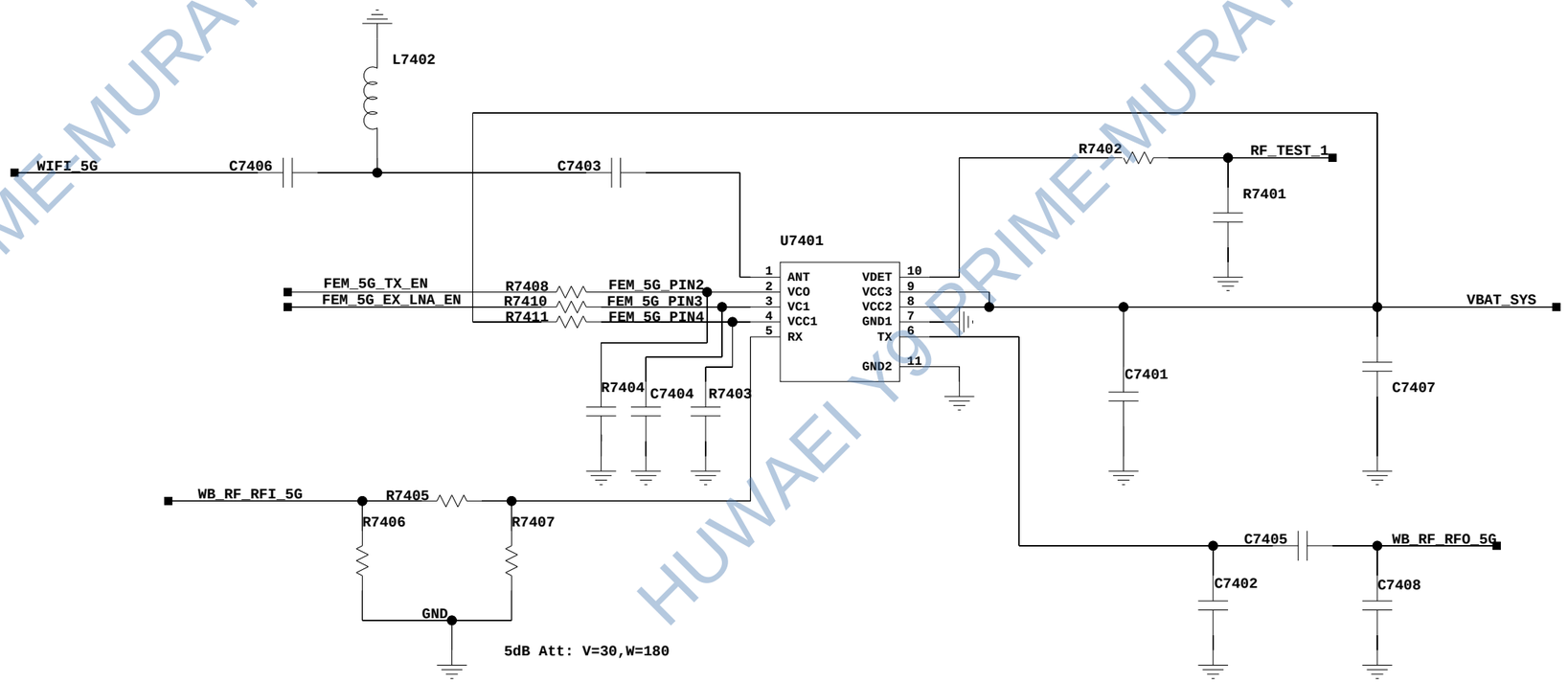
73 NC_2G/5G/GPS

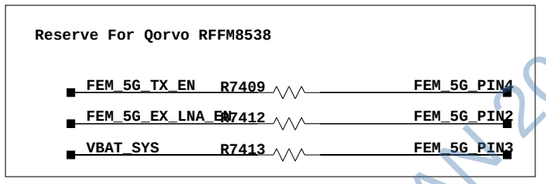


13010676 FOR AL/TL/LX2
13010694 FOR LX1/LX3

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74 NC_FE_WIFI_5G_599





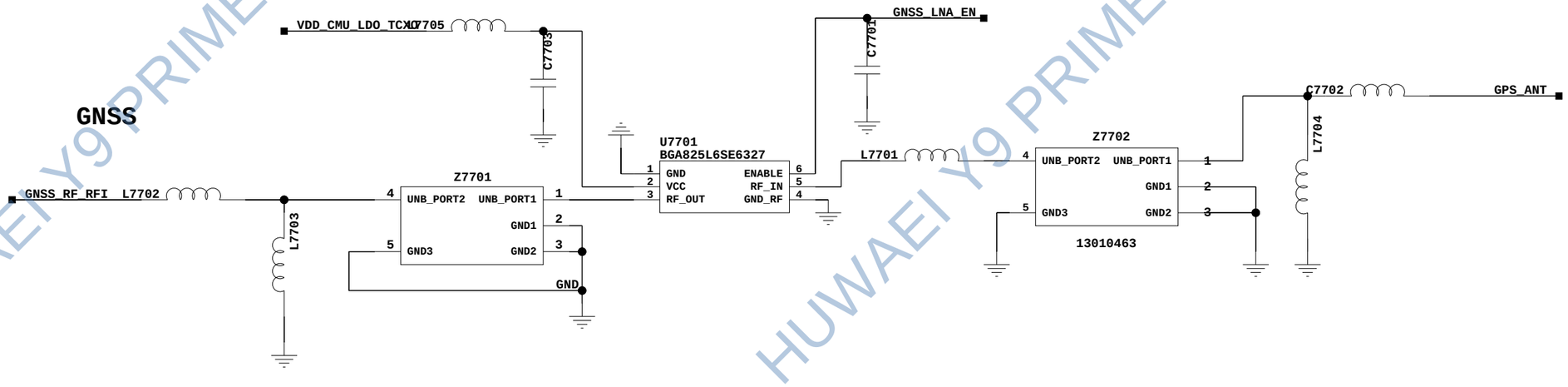
The type and specification of the components refer to the BOM

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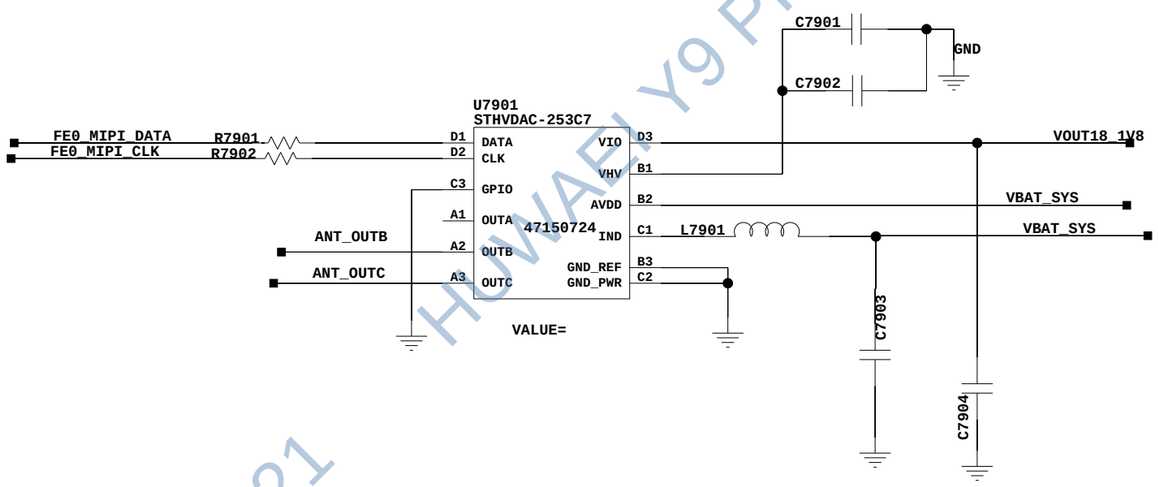
77 NC_GPS



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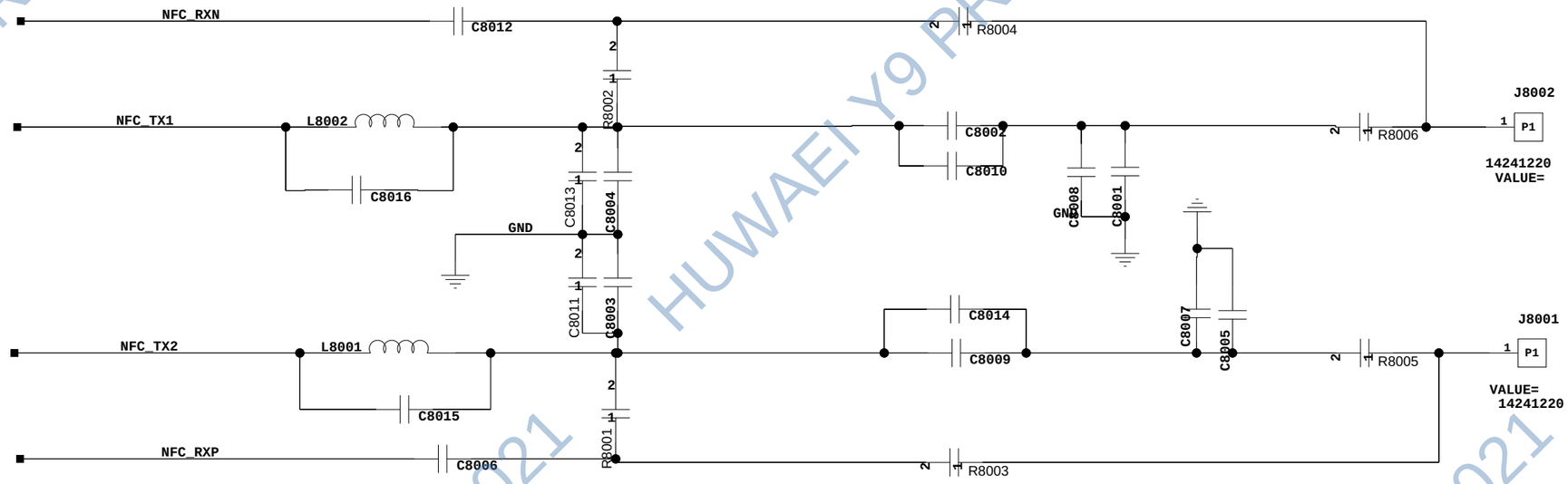
79.ANT_Tuner_controller



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80. ANT_NFC



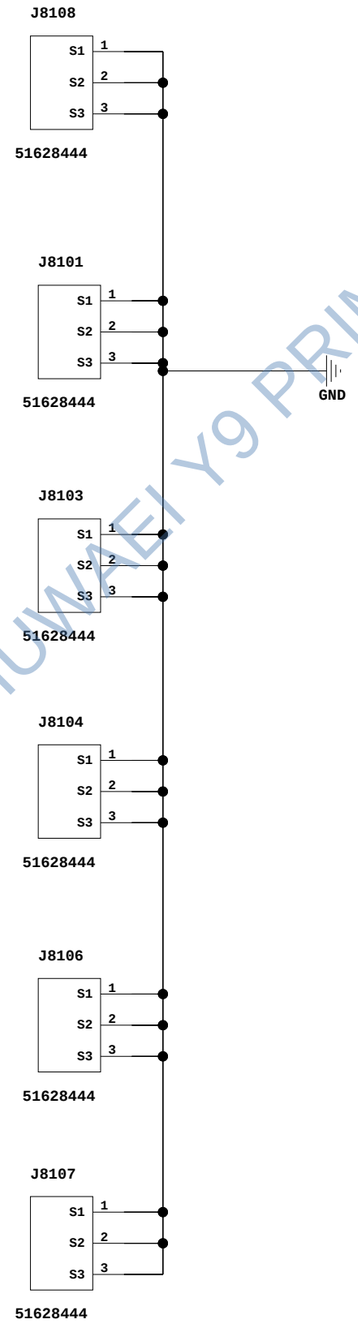
The type and specification of the components refer to the BOM

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81.ANT_GND

div-gnd

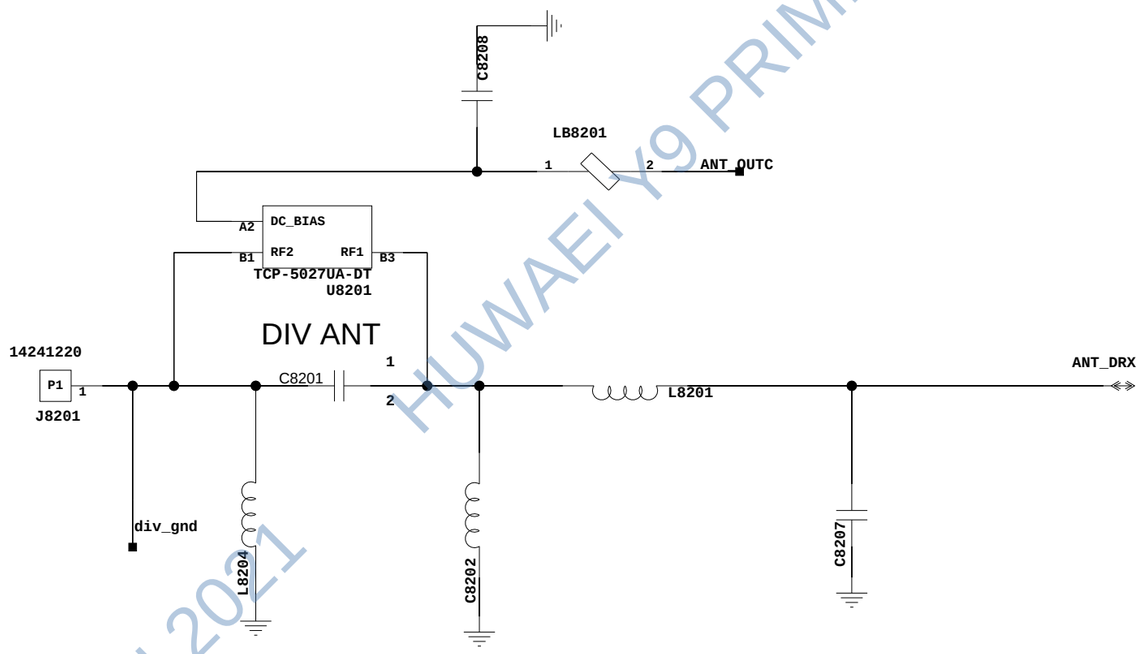
Wifi ground-1



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82.ANT_DIV_feed



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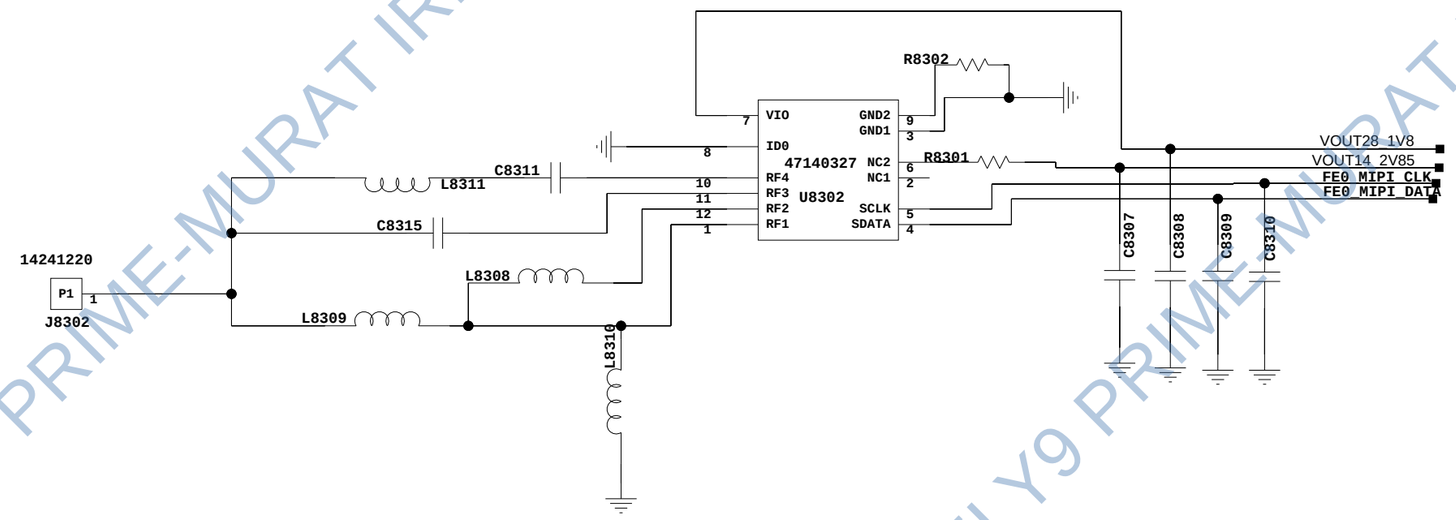
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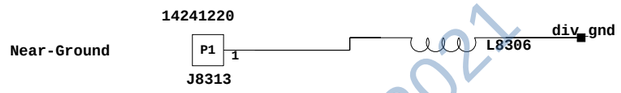
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83.ANT_DIV_tuner

USID Setting Table(Register:0X1F D3:D0)

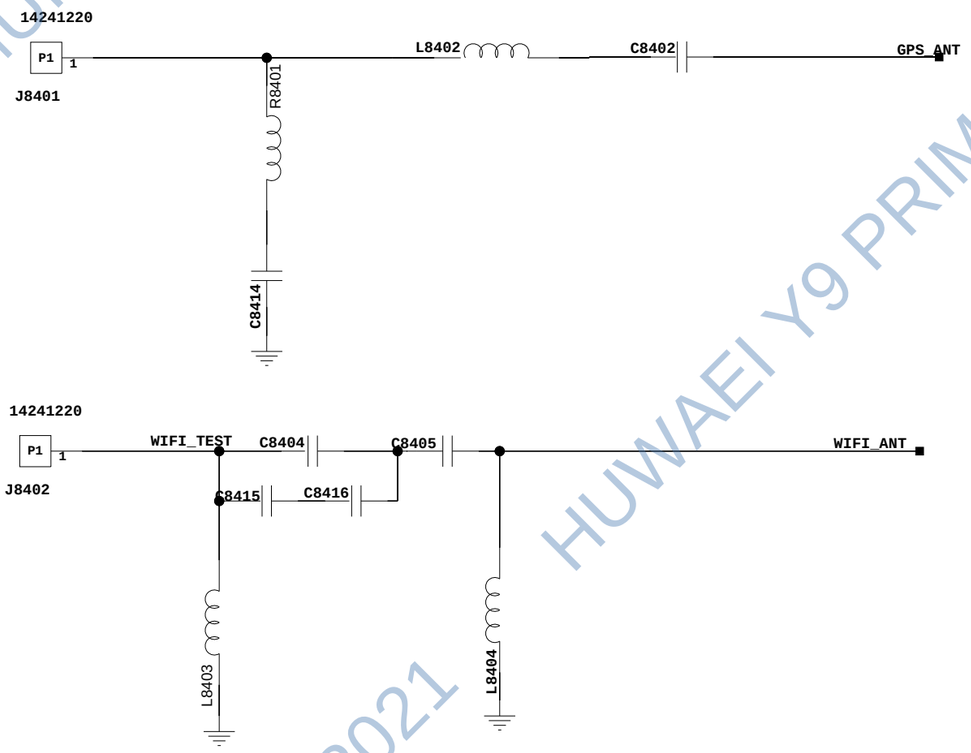
USID PIN BIAS Condition	USID D3	D2	D1	D0
GND	0	1	1	0
Open	0	1	1	1
VIO	1	0	0	1





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84.ANT_GPS_WIFI



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85.ANT_MIMO

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